Chapter 14

Thermal-Network Topology and Quantum Performance Metrics

Where Thermal Signatures Meet Quantum Information

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

After completing this chapter, the reader will be able to:

- **LO-14.1** Explain the bidirectional relationship between thermal engineering and quantum technology: thermal control enables quantum devices, and quantum sensors enable thermal diagnostics.
- **LO-14.2** Identify the critical thermal bottlenecks in superconducting quantum computers, trapped-ion systems, and quantum communication terminals.
- **LO-14.3** Construct a Mode-Path Matrix for a cryogenic quantum system, identifying critical paths and bottlenecks from room temperature to the millikelvin stage.
- **LO-14.4** Design a two-tier thermal monitoring architecture using LWIR system-level imaging and NV-center device-level probing.
- **LO-14.5** Apply the four-axis failure framework (I/M/F/T) to quantum thermal management, with emphasis on cryogenic interface degradation.
- **LO-14.6** Evaluate the mode dominance ratio Γ at cryogenic temperatures and explain why quantum thermal paths are overwhelmingly conduction-dominated.

Abbreviations and Nomenclature

This chapter draws on terminology from quantum information science, cryogenic engineering, and thermal photonics. The following list collects every abbreviation and principal symbol used in Chapter 14 for quick reference.

Acronyms

Abbreviation	Meaning
ALD	Atomic Layer Deposition
BBR	Blackbody Radiation (shift) — systematic frequency shift in optical clocks due to ambient thermal radiation
CP	Cold Plate — intermediate cryostat stage (typically 4 K or 0.7 K)
CVD	Chemical Vapor Deposition
DR	Design Rule (numbered as DR-14Q.n in this chapter)
DR-T	Design Rule — Topological series (Mode-Path Matrix rules from Ch. 4)
EMT	Effective Medium Theory (Ch. 5)

Abbreviation	Meaning
FOV	Field of View
FWHM	Full Width at Half Maximum
I/M/F/T	Interface / Material / Fabrication / Temporal — four-axis failure classification framework (Ch. 12)
LO	Learning Objective (numbered as LO-14.n)
LWIR	Long-Wave Infrared (λ = 8–14 μm); atmospheric transmission window used for Tier-1 thermal monitoring
MPM	Mode-Path Matrix — the book's central thermal network analysis tool (Ch. 4)
MXC	Mixing Chamber — coldest stage of a dilution refrigerator (typically 10–20 mK)
NETD	Noise-Equivalent Temperature Difference — thermal camera sensitivity metric
NV	Nitrogen-Vacancy (center in diamond) — quantum defect used for Tier-2 nanoscale thermometry
ODMR	Optically Detected Magnetic Resonance — readout technique for NV-center spin state
PPKTP	Periodically Poled Potassium Titanyl Phosphate ($KTiOPO_4$) — nonlinear crystal for entangled photon generation
PPLN	Periodically Poled Lithium Niobate ($LiNbO_3$) — nonlinear crystal for frequency conversion and entangled photon sources
PVD	Physical Vapor Deposition
QC	Quantum Computing
QEC	Quantum Error Correction — redundant encoding that detects and corrects qubit errors; thermal crosstalk can defeat QEC if it correlates errors
QKD	Quantum Key Distribution — quantum-secured communication protocol
RF	Radio Frequency
SNSPD	Superconducting Nanowire Single-Photon Detector — cryogenic photon counter for QKD and photonic QC
TPD	Thermal Photonics & Diffusionics — this book's unified framework
TTI	Thermal Trajectory Identification — diagnostic framework linking thermal signatures to quantum performance metrics (introduced in §14.4)
WE	Worked Example (numbered as WE-14.n)

Principal Symbols

Symbol	Meaning	SI Unit
Thermal quantities		
Γ	Mode dominance ratio, $\Gamma \equiv G_{\text{rad}}/G_{\text{cond}}$	—
G_{cond}	Thermal conductance (conduction path)	$\mathrm{W\,K^{-1}}$
G_{rad}	Thermal conductance (radiation path), linearized as $4\varepsilon\sigma_{\text{SB}}T^3A$	$\mathrm{W\,K^{-1}}$
κ	Thermal conductivity (scalar or isotropic)	$\mathrm{W\,m^{-1}\,K^{-1}}$
$\boldsymbol{\kappa}$	Thermal conductivity tensor (anisotropic)	$\mathrm{W\,m^{-1}\,K^{-1}}$

Symbol	Meaning	SI Unit
$\kappa_\parallel$, $\kappa_\perp$	Parallel / perpendicular components of $\boldsymbol{\kappa}$	$\mathrm{W\,m^{-1}\,K^{-1}}$
ε	Total hemispherical emissivity	—
σ_{SB}	Stefan–Boltzmann constant ($5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$)	$\mathrm{W\,m^{-2}\,K^{-4}}$
k_{B}	Boltzmann constant ($1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$)	$\mathrm{J\,K^{-1}}$
R_K	Kapitza (thermal boundary) resistance; dominates cryogenic interfaces	$\mathrm{K\,m^2\,W^{-1}}$
$\dot{Q}$	Heat load (power dissipated or conducted)	W
R_{critical}	Critical-path thermal resistance	$\mathrm{K\,W^{-1}}$
β	Bottleneck ratio — ratio of single-element resistance to total critical-path resistance	—
$\mathbf{q}$	Heat flux vector	$\mathrm{W\,m^{-2}}$
T	Temperature field	K
∇T	Temperature gradient	$\mathrm{K\,m^{-1}}$
τ_{th}	Thermal time constant	s
Quantum quantities		
T_1	Qubit energy relaxation time	s
T_2	Qubit dephasing (coherence) time	s
C_{ij}	Thermal correlation matrix element between qubits i and j; unity = fully correlated, zero = independent	—
d	QEC code distance — determines error-correction capacity; larger d tolerates more errors	—
$d_{\max}$	Maximum useful code distance set by thermal correlation (DR-14Q.3)	—
ρ_{thermal}	Thermal correlation density — fraction of C_{ij} elements exceeding threshold ϵ	—
$\eta_i^{\mathrm{decoherence}}$	Decoherence penalty factor for thermal path i (research hypothesis, §14.6)	—
G_i^{eff}	Effective conductance with quantum penalty	$\mathrm{W\,K^{-1}}$
D	NV-center zero-field splitting frequency (~ 2.87 GHz at 300 K)	Hz
dD/dT	Temperature sensitivity of NV-center ZFS (≈ -74 kHz/K)	$\mathrm{Hz\,K^{-1}}$
Optical / photonic quantities		
λ	Wavelength	m
dn/dT	Thermo-optic coefficient	$\mathrm{K^{-1}}$
α	Optical absorption coefficient	$\mathrm{m^{-1}}$
w_0	Gaussian beam waist radius	m
f_{th}	Thermal lens focal length (in PPLN/PPKTP)	m
$\Delta\nu_{\mathrm{BBR}}$	Blackbody-radiation-induced frequency shift (optical clocks)	Hz

Subscript and Superscript Conventions

Notation	Convention
cond, rad	Conduction and radiation modes, respectively
eff	Effective (combined or penalty-adjusted) value
$\parallel$, $\perp$	Parallel and perpendicular to a reference axis (for anisotropic conductivity)
SB	Stefan–Boltzmann
K	Kapitza (as in R_K, boundary resistance)
th	Thermal (as in τ_{th}, thermal time constant)
proc	Process (as in τ_{proc}, allowable process time)
max	Maximum (as in d_{max})
mount	Mechanical mount conductance path

Design rules for this chapter follow the DR-14Q.n numbering convention. Topological design rules inherited from Chapter 4 retain the DR-T.n prefix. Worked examples use WE-14.n. Learning objectives use LO-14.n.

Chapter Roadmap and Deliverables

This chapter is organized around a single engineering deliverable:

A dual Mode-Path Matrix (cooling vs. parasitic) that converts a quantum system's thermal topology into a monitoring & mitigation plan tied to quantum metrics.

To help you navigate, Figure 14.1 shows the chapter "spine" (workflow) and where the two diagnostic tiers (LWIR system-level and NV device-level) plug into the verification loop.

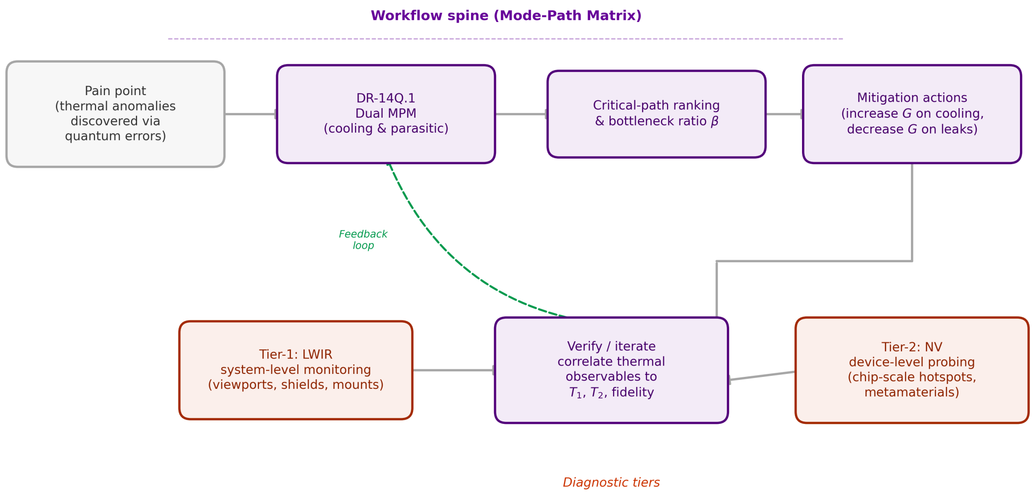

Figure 14.1: Chapter 14 roadmap. The narrative spine is the **dual Mode-Path Matrix** workflow (cooling vs. parasitic paths), which produces a ranked bottleneck list and mitigation actions. The two diagnostic tiers (LWIR and NV) feed the verification loop that ties thermal signatures back to quantum metrics (T_1, T_2, fidelity). The dashed feedback arrow (green) indicates that the verify/iterate step updates the Mode-Path Matrix with measured couplings, closing the thermal–quantum correlation loop (DR-14Q.2).

What To Do First (Reader Recipe)

R-14.1 Build the **cooling MPM** from the 300 K boundary to the device stage; confirm budget margin (WE-14.1).

R-14.2 Build the **parasitic MPM**; rank leaks by β and by coupling-to-qubit relevance (DR-14Q.2).

R-14.3 Attach **Tier-1** sensors for system-level drift, then **Tier-2** sensors for device-level localization.

R-14.4 Close the loop: correlate thermal observables to *measured* T_1, T_2, and gate/logic metrics.

Conduction Mode (Diffusionics)

Dominant mode at cryogenic temperatures: At $T < 4$ K, the Stefan-Boltzmann radiative flux scales as T^4, yielding $G_{\text{rad}} \propto T^3 \to 0$ while Kapitza-regime conduction provides finite G_{cond}. Thus $\Gamma = G_{\text{rad}}/G_{\text{cond}} \ll 0.1$ — the system is overwhelmingly conduction-dominated. The entire cryogenic thermal network lies firmly in the *diffusionics* regime.

Radiation Mode (Thermal Photonics)

Radiation relevance: Despite conduction dominance at mK, radiation plays a critical role in two contexts: (1) *Stray infrared photon loading* from warmer stages (even the 4 K shield emits photons that carry ~ 0.3 meV energy relative to the 15 mK base temperature), and (2) *Diagnostic thermal imaging* of room-temperature and 77 K components using LWIR cameras (Ch. 8 tools). The thermal photonics toolkit thus operates at the boundaries of the quantum system, not in its core.

Multiphysics Integration

Temporal regime: Quantum systems exhibit both steady-state thermal management (continuous cooling of base stage) and transient thermal events (qubit gate pulses, measurement-induced heating, cooldown/warmup cycles). The Mode-Path Matrix must be evaluated at *both* timescales.

Pain Point

Quantum computing and communication systems operate at extreme thermal boundaries — from millikelvin dilution refrigerators to room-temperature fiber-optic terminals. Every thermal anomaly (a warming wirebond, a drifting laser mount, a degrading cryogenic interface) directly degrades quantum coherence, gate fidelity, or key generation rate. Yet thermal diagnostics in quantum systems remain *ad hoc*: no systematic framework connects thermal network topology to quantum performance metrics. The result is that thermal failures are discovered through quantum errors, not through thermal monitoring — making root-cause analysis slow and expensive.

14.1 The Bidirectional Framework

The relationship between thermal engineering and quantum technology is fundamentally **bidirectional**:

1. **Thermal → Quantum:** Precision thermal management (temperature stability, gradient control, heat extraction) is a prerequisite for quantum device operation. Every millikelvin of temperature instability, every microwatt of parasitic heat load, directly degrades quantum coherence.

2. **Quantum → Thermal:** Quantum sensing technologies — particularly nitrogen-vacancy (NV) centers in diamond — offer thermal measurement capabilities that exceed classical instruments by orders of magnitude in spatial resolution, enabling nanoscale thermal diagnostics of metamaterial devices.

This bidirectional relationship creates a *virtuous cycle*: better thermal engineering improves quantum devices, which in turn provide better thermal diagnostic tools for the next generation of thermal engineering. Figure 14.2 illustrates this cycle and maps the key technology linkages.

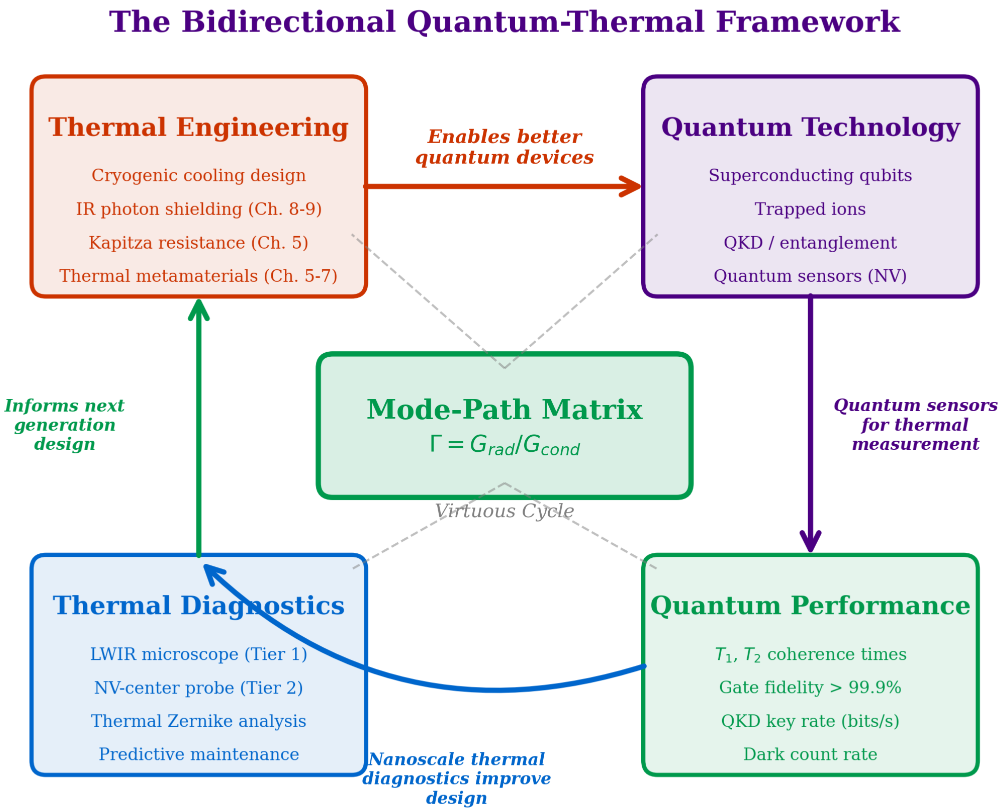

Figure 14.2: The bidirectional quantum–thermal framework. **Left:** thermal engineering enables quantum devices through precision temperature control, heat extraction, and gradient management. **Right:** quantum sensors (NV centers, superconducting thermometers) provide unprecedented thermal diagnostic resolution, closing the design loop. The virtuous cycle connects the Mode-Path Matrix methodology (Ch. 4) to quantum performance metrics (T_1, T_2, gate fidelity).

Trilogy Connection:

Fluctuation–Dissipation Connection The mode dominance ratio $\Gamma = G_{\text{rad}}/G_{\text{cond}}$ quantifies the competition between coherent (photon) and incoherent (phonon) energy transport. This is, at its deepest level, the same competition that governs quantum decoherence: thermal photons destroy coherence, while phonon-mediated processes thermalize quantum states. The Mode-Path Matrix thus provides a thermal engineer's perspective on the decoherence budget — a concept developed from the quantum optics side in *Quantum Field Imaging* (Book 2, Ch. 8).
The fluctuation–dissipation theorem, introduced in Chapter 3 for near-field heat transfer, takes on new meaning here: every dissipative thermal pathway is also a fluctuation source that injects noise into quantum systems. Minimizing the thermal network's dissipation simultaneously minimizes quantum noise.

14.1.1 Mathematical Statement

For a quantum device with coherence time T_1 operating at temperature T_{base} with a thermal environment characterized by Mode-Path Matrix $\{\Gamma_i\}$, the thermal contribution to decoherence can be bounded by:

$$\frac{1}{T_1^{\text{thermal}}} \leq \sum_{\text{paths } i} \frac{k_{\text{B}}\,\delta T_i}{\hbar\,\omega_q} \cdot g_i \tag{14.1}$$

where ω_q is the qubit transition frequency, δT_i is the temperature fluctuation along path i, and g_i is the coupling strength between path i and the qubit degree of freedom. The summation runs over *all* paths in the thermal network — precisely those identified by the Mode-Path Matrix.

Quantum-Thermal Insight:

Temperature Fluctuations as Decoherence Source Equation (14.1) reveals why the Mode-Path Matrix is valuable for quantum systems: it systematically identifies *all* thermal pathways that could inject noise. Missing a thermal path (violating DR-T.1) means missing a decoherence channel. The critical path in the thermal network is often also the dominant decoherence pathway.

14.1.2 The Quantum Thermal Management Challenge

Unlike semiconductor fabrication, where the goal is typically to *transport* heat efficiently (minimize thermal resistance), quantum systems present an **inverted design problem**:

- **Semiconductor fab:** Minimize R_{critical} (maximize heat extraction).
- **Quantum systems:** Maximize isolation resistance on parasitic paths while maintaining low resistance on intentional cooling paths.

This means the Mode-Path Matrix workflow (Section 4.6) operates in a *dual mode* for quantum systems:

Table 14.3: Inverted design objectives: semiconductor vs. quantum thermal management.

Design Aspect	**Semiconductor (Ch. 11)**	**Quantum (Ch. 14)**
Critical path goal	Minimize R	Minimize R on cooling path; maximize R on parasitic paths
Bottleneck action	Increase G	Increase G (cooling) or decrease G (parasitic)
Γ preference	Context-dependent	$\Gamma \to 0$ at device stage
Thermal metamaterial role	Concentrators, via arrays	Thermal isolators, radiation shields
Failure mode	Overheating	Excess heat load $\to$ decoherence

Design Rule:

DR-14Q.1 — Dual Network Optimization For quantum thermal systems, construct *two* Mode-Path Matrices: one for intentional cooling paths (minimize R_{critical}) and one for parasitic heat leak paths (maximize $R_{\text{parasitic}}$). Optimize both simultaneously. A design that improves cooling but inadvertently creates a parasitic thermal short will degrade quantum performance.

WE-14.1: Cryostat Mode-Path Matrix

Problem: A dilution refrigerator for a 100-qubit superconducting quantum computer has the following parameters: MXC cooling power: 20 μW at 20 mK; DC wiring: 200 phosphor-bronze lines, each $\kappa L/A = 3 \times 10^{-7}$ W/K from 4 K to MXC; RF wiring: 100 NbTi coax lines, each $G_{\text{cond}} = 5 \times 10^{-8}$ W/K from 4 K to MXC; Radiation load from 4 K shield: $\dot{Q}_{\text{rad}} = 0.5$ μW; Kapitza resistance at MXC–chip interface: $R_K = 500$ K/W at 20 mK.

Find: (a) Total heat load on MXC. (b) Critical parasitic path. (c) Bottleneck ratio β. (d) Cooling budget sufficiency. (e) Mode classification.

Solution:

(a) DC wiring load to MXC (thermalized at cold plate, $\Delta T = 0.08$ K):

$$\dot{Q}^{\text{DC,MXC}} = 200 \times 3 \times 10^{-7} \times 0.08 = 4.8\ \mu\text{W} \tag{14.2}$$

RF wiring load:

$$\dot{Q}^{\text{RF,MXC}} = 100 \times 5 \times 10^{-8} \times 0.08 = 0.4\ \mu\text{W} \tag{14.3}$$

Total: $\dot{Q}^{\text{total}} = 4.8 + 0.4 + 0.5 = 5.7$ μW.

(b) DC wiring contributes $4.8/5.7 = 84\%$ — **critical parasitic path.**

(c) $\beta = 0.84 > 0.5$ — single dominant bottleneck.

(d) 5.7 μW $<$ 20 μW: **sufficient** (71% margin). Loaded base temperature:

$$T_{\text{base}} \approx 15 \times \sqrt{1 + 5.7/20} \approx 17\text{ mK} < 20\text{ mK} \quad \checkmark \tag{14.4}$$

(e) All paths $\Gamma < 10^{-6}$ except 4 K radiation ($\Gamma \gg 10$).

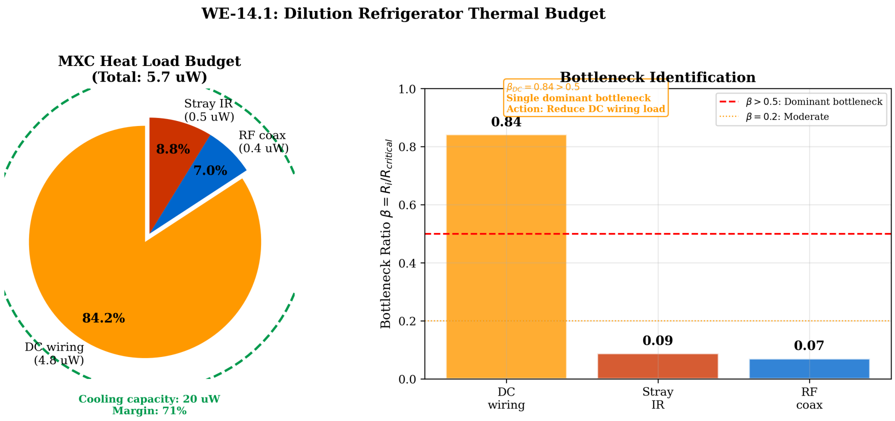

Figure 14.3: Bottleneck analysis for WE-14.1: heat load budget at the mixing chamber stage. DC wiring dominates at 84% ($\beta = 0.84$), with RF wiring (7%) and stray radiation (9%). Total load (5.7 μW) within 20 μW budget (71% margin), yielding $T_{\text{base}} \approx 17$ mK.

$$\boxed{T_{\text{base}} \approx 17 \text{ mK}; \quad \beta_{\text{DC wiring}} = 0.84; \quad \text{Budget margin: } 71\%} \tag{14.5}$$

14.2 Critical Thermal Bottlenecks in Quantum Systems

We now apply the Mode-Path Matrix methodology (DR-T.1 through DR-T.4) to survey the thermal pain points across major quantum technology platforms.

14.2.1 Superconducting Quantum Computers

Superconducting qubits (transmons, flux qubits) operate at $T \approx$ 10–20 mK inside dilution refrigerators. The thermal network spans five orders of magnitude in temperature: from 300 K (room-temperature electronics) to $\sim$15 mK (mixing chamber).

Table 14.4: Thermal stages of a typical dilution refrigerator with thermal budget.

Stage	T (K)	**Cooling Power**	**Wiring Load**	Γ	**Bottleneck?**
Room temp. plate	300	—	—	$\sim$1	No
50 K stage	50	40 W	10 W (DC)	10^{-2}	Rarely
4 K stage	4	1.5 W	500 mW (RF)	10^{-4}	Sometimes
Still	0.8	20 mW	50 mW	10^{-6}	Often
Cold plate (CP)	0.1	500 μW	10 μW	10^{-8}	Yes
Mixing chamber (MXC)	0.015	20 μW	5 μW	10^{-10}	**Critical**

The critical thermal bottlenecks are:

1. **Wiring thermal load:** Coaxial cables and DC lines carry heat from warmer stages. Each RF coax contributes $\sim$1 μW to the MXC stage. For a 1000-qubit system requiring $\sim$2000 signal lines, the total wiring load approaches or exceeds the MXC cooling power.

2. **Cryogenic thermal interfaces:** Every bolted joint in the cryostat introduces a Kapitza resistance. At $T < 1$ K, Kapitza resistance scales as $R_K \propto T^{-3}$, making interfaces increasingly resistive as temperature drops.

3. **Infrared photon loading:** Despite radiation shields, stray IR photons from the 4 K stage (peak emission at $\lambda_{\text{peak}} \approx 700\ \mu$m) can reach the MXC. Even sub-nanowatt radiative loads are significant when cooling power is only 20 μW.

Failure Mode:

Quasiparticle Poisoning via Stray Photons A single infrared photon at $\lambda = 700\ \mu$m carries energy $E = hc/\lambda \approx 0.3$ meV. In a superconducting aluminum qubit with gap $\Delta \approx 0.18$ meV, this photon can break Cooper pairs and generate quasiparticles, increasing the quasiparticle density n_{qp} and directly reducing T_1. This is a *radiation-mode failure* in what appears to be a purely conduction-dominated system ($\Gamma \sim 10^{-10}$): the tiny radiative conductance carries disproportionate impact because its "payload" (broken Cooper pairs) has outsized quantum consequences.

Thermal photonics connection: The radiation shielding techniques of Chapter 9 (spectral selectivity, multiphysics cloaking) apply directly to IR photon blocking in dilution refrigerators. Low-pass IR filters at each cryostat stage are, in effect, *cold mirrors* (Ch. 8) optimized for far-infrared rejection.

14.2.2 Trapped-Ion Quantum Computers

Trapped-ion systems operate at or above room temperature for the ion trap itself but require extreme thermal stability:

Table 14.5: Thermal pain points in trapped-ion quantum systems.

Component	Thermal Issue	Γ	Mode-Path Matrix Role
Trap electrodes	Laser-induced heating; anomalous motional heating	~ 0.1–1	Mixed-mode path (Ch. 9)
Vacuum chamber window	Thermal lensing from laser absorption	~ 1	Radiation path; dn/dT analysis (Ch. 8)
Imaging optics	Thermal drift affects ion-to-camera registration	~ 0.5	Conduction + radiation
RF electronics	Dissipation heats trap mount	< 0.1	Conduction-dominated

Quantum-Thermal Insight:

Anomalous Heating as Thermal Signature Trapped ions exhibit "anomalous motional heating" — an unexplained heating of the ion's secular motion at rates far exceeding predictions from Johnson noise. The electric field noise spectral density scales as $S_E \propto d^{-4}$ where d is the ion–electrode distance, suggesting a surface phenomenon. One hypothesis is that thermal fluctuations at the electrode surface drive adsorbed contaminant dipoles. If correct, the Mode-Path Matrix for the electrode thermal network — specifically, the surface temperature fluctuation spectrum — could predict anomalous heating rates. This remains an open research question with high potential impact.

14.2.3 Quantum Communication Systems

Quantum key distribution (QKD) and entanglement distribution systems face thermal challenges at both the source and detector ends:

Table 14.6: Thermal challenges in quantum communication systems.

Component	Thermal Sensitivity	Requirement	Γ
PPLN/PPKTP crystal	Phase-matching $\propto T$	$\Delta T < \pm 5$ mK	< 0.01
Laser diode (pump)	Wavelength $\propto T$	$\Delta T < \pm 10$ mK	< 0.1
SNSPD detector	Detection efficiency $\propto$ bias; thermal crosstalk between pixels	$\Delta T < \pm 50$ mK	$< 10^{-6}$
Fiber coupling	Alignment drift $\propto \nabla T$	$\nabla T < 0.1$ K/m	< 0.01

The nonlinear crystals (PPLN, PPKTP) used for spontaneous parametric down-conversion (SPDC) or frequency conversion have phase-matching bandwidths of order 0.1–1 nm, corresponding to temperature tolerances of 5–50 mK depending on crystal length and material.

14.3 Two-Tier Thermal Monitoring Architecture

We propose a **two-tier monitoring architecture** that provides both system-level overview and device-level precision, as illustrated in Figure 14.4.

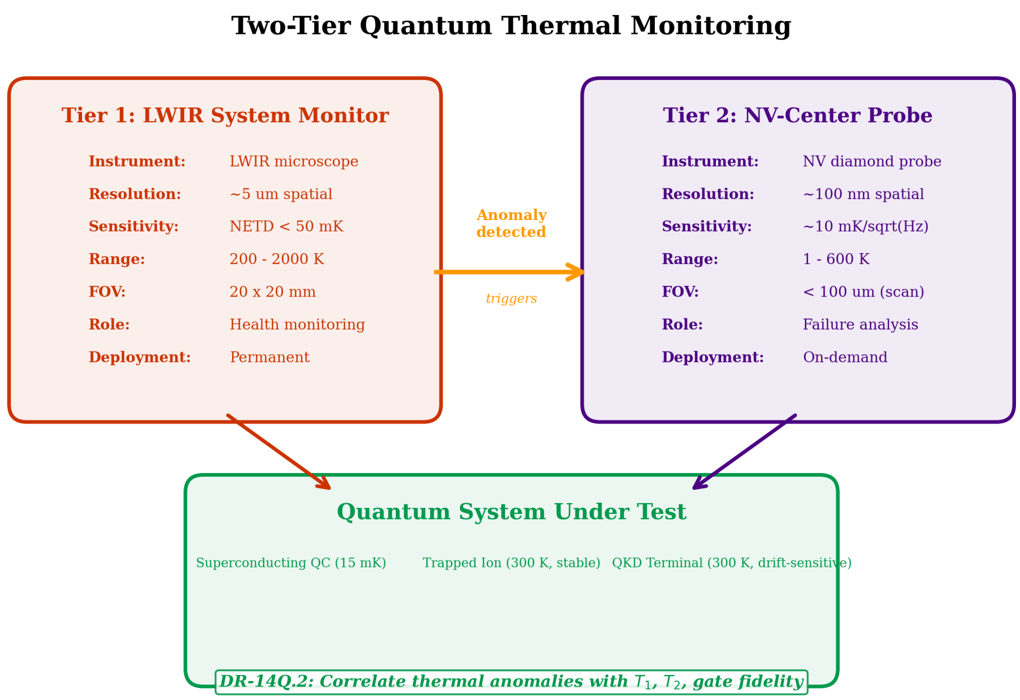

Figure 14.4: Two-tier quantum thermal monitoring architecture. **Tier 1** (left): LWIR microscope module provides system-level full-field imaging at ~ 5 μm resolution for operational health monitoring and anomaly detection. **Tier 2** (right): NV-center scanning thermal probe delivers device-level point measurements at $\sim$100 nm resolution for failure analysis and physics exploration. The two tiers are complementary: Tier 1 identifies regions of interest; Tier 2 investigates them at nanoscale.

Table 14.7: Two-tier quantum thermal monitoring architecture.

	Tier 1: System Level	**Tier 2: Device Level**
Instrument	LWIR microscope module	NV-center thermal probe
Spatial resolution	$\sim 5\ \mu$m	~ 100 nm
Temperature range	200–2000 K	1–600 K
Sensitivity	NETD $\sim$ 20–50 mK	~ 10 mK/$\sqrt{\text{Hz}}$
Field of view	10–100 mm	$< 100\ \mu$m (scanning)
Measurement mode	Full-field imaging	Point or raster scan
Primary role	Operational health monitoring; anomaly detection	Failure analysis; physics exploration
Deployment	Permanent installation	Brought in for diagnosis
Relevant chapter	Ch. 8 (spectral engineering)	Ch. 3 (near-field), Ch. 5 (NV physics)

14.3.1 Tier 1: LWIR Microscope Module

The system-level thermal monitor is an LWIR (long-wave infrared, λ = 8–14 μm) microscope with design specifications given in Table 14.8.

Table 14.8: LWIR microscope module specifications for quantum system monitoring.

Parameter	**Specification**	**Rationale**	**Design Driver**
Spectral range	8–14 μm	Atmospheric window; peak emission at 300 K	Detector selection (microbolometer)
NETD	< 50 mK	Detect ± 0.1 K anomalies	Integration time
Spatial resolution	$\leq 5\ \mu$m	Resolve bond wires, solder joints	Lens NA; diffraction limit $\sim 5\ \mu$m
Working distance	50–200 mm	Access through cryostat viewports	IR window material (ZnSe, Ge)
Frame rate	≥ 30 Hz	Track transient events	Detector readout
FOV	$\geq 20 \times 20$ mm	Cover package or cryostat stage	Sensor array size

Trilogy Connection:

LWIR as Thermal Wavefront Sensor Drawing on concepts from *The Eikonal Bridge* (Book 1), the LWIR microscope can be reframed as a **thermal wavefront sensor**. Just as a Shack-Hartmann sensor decomposes an optical wavefront into aberration modes (Zernike polynomials), a thermal image can be decomposed into a basis of spatial thermal modes: Piston (Z_0): uniform offset $\rightarrow$ global cooling deficiency; Tilt (Z_1, Z_2): linear gradient $\rightarrow$ asymmetric heat load; Defocus (Z_4): radial curvature $\rightarrow$ center-to-edge imbalance; Coma (Z_7, Z_8): asymmetric $\rightarrow$ single-point heat source; High-order: fine structure $\rightarrow$ local defects. By monitoring the time evolution of thermal Zernike coefficients, operators can diagnose failure modes before they impact quantum performance.

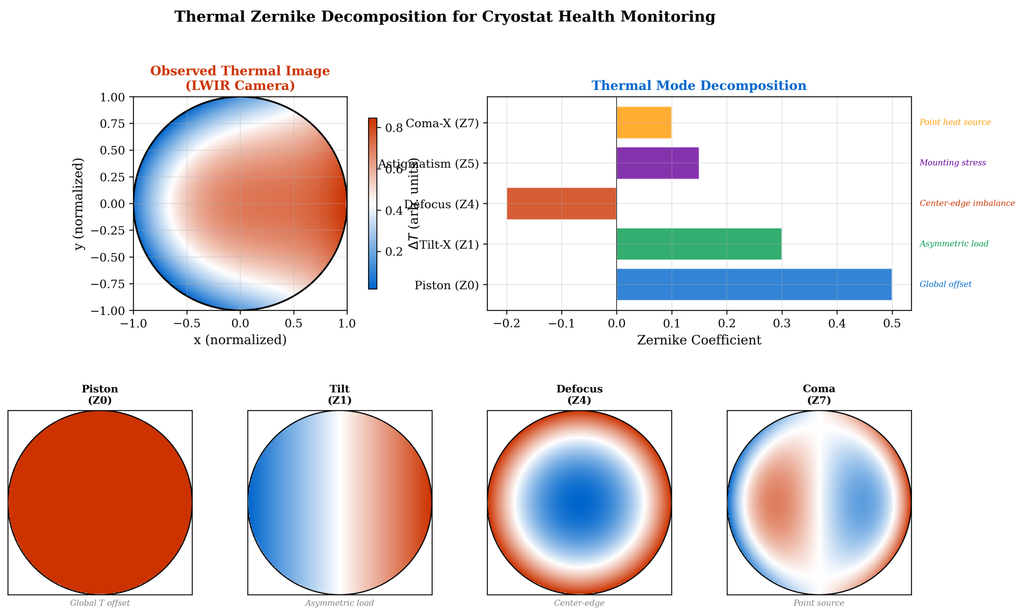

Figure 14.5: Thermal Zernike decomposition applied to a cryostat radiation shield. **Top row:** measured LWIR thermal image and its decomposition into the first 15 Zernike modes. **Bottom row:** individual mode contributions — piston (global offset), tilt (asymmetric load), defocus (radial gradient), and coma (localized hot spot). Temporal tracking of Zernike coefficients enables predictive maintenance.

14.3.2 Tier 2: NV-Center Thermal Probe

The nitrogen-vacancy center in diamond provides nanoscale thermometry through the temperature dependence of its optically detected magnetic resonance (ODMR) spectrum.

Physical mechanism: The NV center's ground-state zero-field splitting $D \approx 2.87$ GHz shifts with temperature at a rate $dD/dT \approx -74$ kHz/K near room temperature. By tracking D via ODMR, temperature is measured with nanometer-scale spatial resolution.

Table 14.9: NV-center thermal probe performance benchmarks.

Parameter	**Single NV**	**Ensemble (nano-diamond)**	**Units**
Spatial resolution	$\sim$ 10 nm (STED)	$\sim$ 100 nm	—
Temperature sensitivity	$\sim$ 100	$\sim$ 10	$\mathrm{mK}/\sqrt{\mathrm{Hz}}$
Dynamic range	1–600	1–600	K
Measurement bandwidth	$\sim$ 100	$\sim$ 1000	Hz
Integration time for 1 mK	$\sim 10^4$	$\sim 10^2$	s

Fabrication Challenge

NV-center thermometry faces practical challenges: (1) the dD/dT coefficient decreases at cryogenic temperatures, reducing sensitivity; (2) optical access to cryogenic stages requires free-space or fiber paths through multiple thermal shields; (3) laser illumination ($\sim$ 100 μW at 532 nm) deposits heat at the measurement point, creating a systematic bias.

14.4 Application: Superconducting Quantum Computers

Platform template quick-view

Nodes (what to model) Cryostat stages (300 K → 50 K → 4 K → Still → MXC); qubit chip/package; cabling (coax/DC/attenuators); readout chain (HEMT, isolators); radiation shields.

Dominant paths (what couples to what) Conduction through cabling and supports; radiative coupling via apertures; microwave dissipation near the chip; substrate/packaging phonon transport; quasiparticle generation routes.

Γ regime (why this platform sits here) Typically deep cryogenic conduction-dominated ($\Gamma \ll 1$); most parasitics behave as $\Gamma \lesssim 10^{-3}$, with localized hotspots effectively $\Gamma \sim 10^{-2}$.

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

We now apply the full Mode-Path Matrix workflow (Section 4.6) to the thermal management of a superconducting quantum computing system.

14.4.1 Step 1: Define Thermal Requirements

Table 14.10: Thermal requirements for a 1000-qubit superconducting quantum computer.

Requirement	**Value**	**Units**	**Consequence of Violation**
Base temperature	< 20	mK	Thermal population of qubit states
Temperature stability	± 1	mK	Qubit frequency drift
MXC cooling power	> 20	μW	Cannot absorb heat loads
Total MXC heat load	< 15	μW	Base temperature rises above 20 mK
Wiring heat load (per line)	< 5	nW	Exceeds cooling budget at 1000+ lines
IR photon flux at MXC	$< 10^3$	photons/s	Quasiparticle poisoning

14.4.2 Step 2: Construct Thermal Network Graph

The dilution refrigerator thermal network has a predominantly **hierarchical tree** topology (Table 4.1), with the room-temperature plate as root and the mixing chamber as the deepest leaf.

Table 14.11: Thermal network nodes for dilution refrigerator system.

Node	Name	T (K)	C (J/K)	Type
N1	Room-temp plate	300	Large	Source
N2	50 K stage	50	200	Junction
N3	4 K stage	4	50	Junction
N4	Still	0.8	5	Junction
N5	Cold plate	0.1	0.5	Junction
N6	Mixing chamber (MXC)	0.015	0.05	Junction
N7	Qubit chip	0.015	0.001	Sink (target)
N8	Wiring harness (DC)	Gradient	0.1	Distributed
N9	Wiring harness (RF)	Gradient	0.05	Distributed
N10	Radiation shield (4 K)	4	100	Shield
N11	Radiation shield (50 K)	50	500	Shield
N12	He-3/He-4 circuit	Varies	—	Active cooling

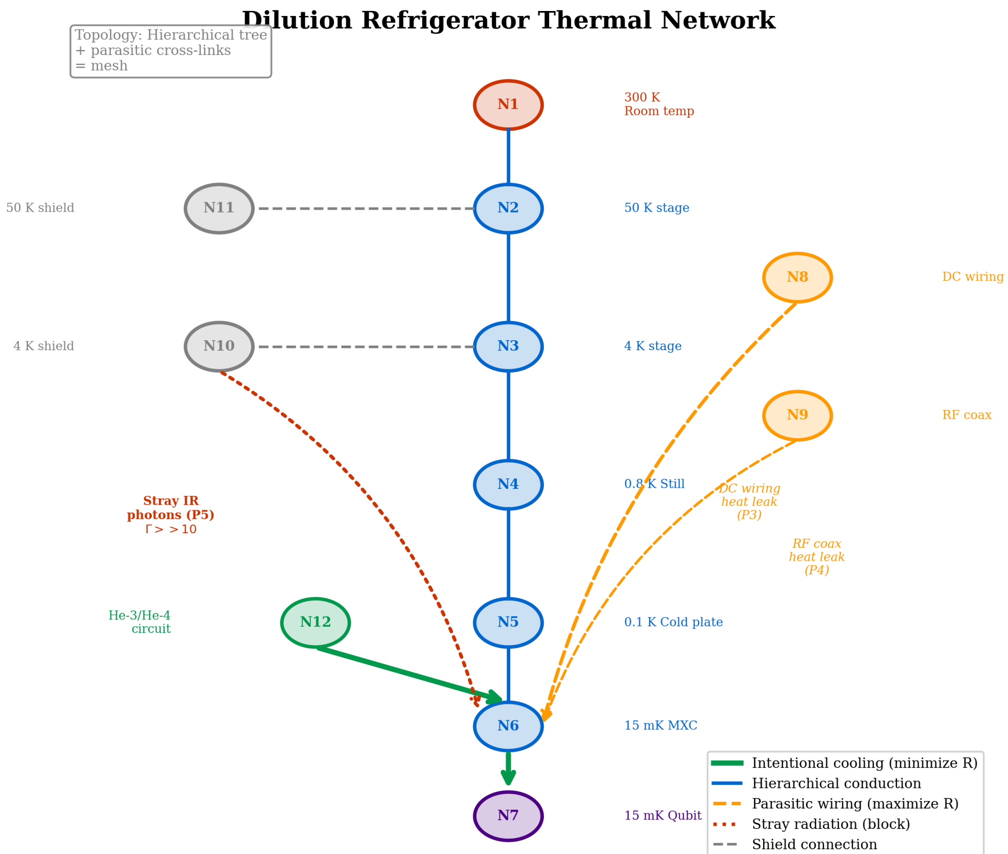

Figure 14.6: Thermal network graph of a dilution refrigerator quantum computing system. Nodes (N1–N12) correspond to thermal stages and components listed in Table 14.11. Solid blue edges: intentional cooling paths (minimize R). Dashed red edges: parasitic heat leak paths (maximize R). The hierarchical tree structure from N1 (300 K) to N7 (15 mK) spans five orders of magnitude in temperature.

14.4.3 Step 3: Analyze Topology and Find Critical Path

The dominant topology is **hierarchical series** from N1 down to N7, with **parallel parasitic paths** (radiation, residual gas) creating cross-connections.

14.4.4 Step 4: Build Mode-Path Matrix

The complete Mode-Path Matrix for the dilution refrigerator system is presented in Table 14.12 and visualized as a heatmap in Figure 14.7.

Table 14.12: Mode-Path Matrix for dilution refrigerator quantum computer.

Path	Route	G_{cond} (W/K)	G_{rad} (W/K)	Γ	Mode	Action
P1	N12 → N6 (He cooling)	$\sim$0.01	$\sim$0	$< 10^{-10}$	Cond	Maximize G
P2	N6 → N7 (MXC–chip)	10^{-4}	$\sim$0	$< 10^{-12}$	Cond	Minimize R_K
P3	N1 → N6 (DC wiring)	5×10^{-6}	$\sim$0	$< 10^{-10}$	Cond	Maximize R
P4	N1 → N6 (RF coax)	2×10^{-6}	$\sim$0	$< 10^{-10}$	Cond	Maximize R
P5	N10 → N6 (stray IR)	$\sim$0	10^{-10}	$\gg 10$	Rad	Block (Ch. 8)
P6	N3 → N6 (residual gas)	10^{-8}	$\sim$0	$< 10^{-6}$	Cond	Improve vacuum

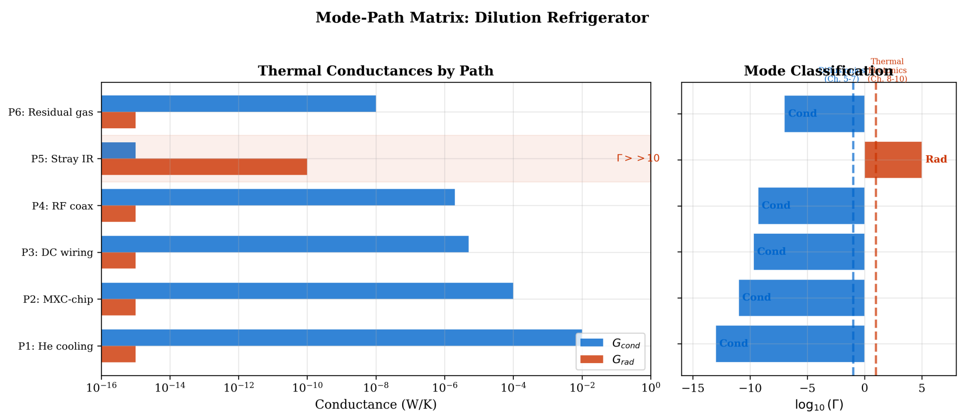

Figure 14.7: Mode-Path Matrix heatmap for the dilution refrigerator quantum computer. Each cell represents a thermal path between node pairs, color-coded by mode dominance: blue ($\Gamma < 0.1$, conduction-dominated), purple ($0.1 < \Gamma < 10$, mixed), and red ($\Gamma > 10$, radiation-dominated). Path P5 (4 K shield → MXC, stray IR) is the sole radiation-dominated path in the entire network.

Quantum-Thermal Insight:

The Stray Radiation Paradox Path P5 has $\Gamma \gg 10$ — it is the *only* radiation-dominated path in the entire cryostat network. Its absolute conductance ($\sim 10^{-10}$ W/K) is tiny, yet it carries disproportionate quantum impact because each absorbed IR photon can break Cooper pairs. This illustrates a key principle: Γ **classifies the physics, not the importance.**

14.4.5 Step 5: Apply Mode-Appropriate Design Tools

Conduction Mode (Diffusionics)

Paths P1–P4, P6 (Conduction-dominated, $\Gamma < 0.1$): P2 (MXC–chip interface): Kapitza resistance management via gold plating or indium press contacts. P3, P4 (Wiring): Superconducting NbTi coax with thermal anchoring at each stage. P6 (Residual gas): Maintain vacuum $< 10^{-6}$ mbar.

Radiation Mode (Thermal Photonics)

Path P5 (Radiation-dominated, $\Gamma > 10$): IR photon blocking requires multi-stage radiation shields with low-emissivity ($\varepsilon < 0.02$) inner surfaces, low-pass IR filters at each stage, and light-tight construction. These are direct applications of the cold mirror (Ch. 8) and radiative shielding (Ch. 9) techniques.

14.4.6 Step 6: Verify and Iterate

After implementing thermal improvements, measure base temperature and correlate with quantum metrics (T_1, T_2, gate fidelity).

Design Rule:

DR-14Q.2 — Thermal–Quantum Correlation Always correlate thermal measurements with quantum performance metrics (T_1, T_2, gate fidelity, error rate). A thermal anomaly that does not affect quantum performance may be benign; a quantum degradation without thermal anomaly points to a non-thermal root cause.

14.5 Application: Quantum Communication Systems

Platform template quick-view

Nodes (what to model) Nonlinear crystal (PPLN/PPKTP); ovens/TEC; pump lasers; fiber coupling; mounts; temperature sensors and controllers.

Dominant paths (what couples to what) Absorption heating in crystal; conduction to oven; air/convection (if not in vacuum); thermo-optic phase-matching drift.

Γ regime (why this platform sits here) Quasi-1D conduction with strong source; $\Gamma \sim 10^{-2}$–10^{-1} at device scale (controller bandwidth matters).

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

14.5.1 QKD Terminal Thermal Monitoring

Table 14.13: Mode-Path Matrix for QKD transmitter terminal.

Path	**From → To**	G_{cond} (W/K)	G_{rad} (W/K)	Γ	**Mode**
P1	Pump laser → TEC	5	0.05	0.01	Cond
P2	PPLN crystal → oven	0.5	0.01	0.02	Cond
P3	Oven → enclosure	0.1	0.3	3.0	Mixed
P4	Enclosure → ambient	0.5	2	4.0	Mixed
P5	Fiber coupling → baseplate	2	0.02	0.01	Cond

14.5.2 SNSPD Thermal Crosstalk

Superconducting nanowire single-photon detectors (SNSPDs) at $T \sim 2$ K on a shared substrate exhibit thermal crosstalk modeled via Mode-Path Matrix with pixel–pixel conduction and negligible radiation ($\Gamma < 10^{-8}$). Thermal isolation trenches (transformation thermotics, Ch. 5) reduce inter-pixel G_{cond} by 10×–100×.

14.6 Mode-Path Matrix for Quantum Systems: Special Considerations

14.6.1 Ultra-Low Temperature Regime

Table 14.14: Thermal physics regime changes at cryogenic temperatures.

Property	**Classical ($T > 50$ K)**	**Cryogenic ($T < 1$ K)**
Thermal conductivity	$\kappa \sim$ const or $\propto 1/T$	$\kappa \propto T^n$ ($n = 1$–3)
Specific heat	Dulong-Petit ($3R$)	$C \propto T^3$ (Debye) or $\propto T$ (electronic)
Interface resistance	$R_c \sim$ const	$R_K \propto T^{-3}$ (Kapitza)
Radiation	$q \propto T^4$ (significant)	$q \propto T^4 \to 0$ (negligible)
Γ	10^{-2} to 10^2	$\ll 10^{-6}$ (always conduction)

14.6.2 Quantum Penalty Extension (Research Direction)

Quantum-Thermal Insight:

Toward a Quantum-Extended Mode-Path Matrix We hypothesize that the Mode-Path Matrix can be extended to include a **quantum penalty** term:

$$G_i^{\text{eff}} = G_i \times \left(1 - \eta_i^{\text{decoherence}}\right) \tag{14.6}$$

where $\eta_i^{\text{decoherence}} \in [0, 1]$ represents the fraction of thermal noise on path i that contributes to qubit decoherence. **Status:** This is a research hypothesis, not established methodology. Falsification criterion: if thermal path-specific perturbations do not produce measurable, path-specific T_1 changes, the model is inadequate.

14.6.3 Metamaterial Solutions for Quantum Thermal Isolation

Table 14.15: Metamaterial toolkit applied to quantum thermal isolation (inverted objectives).

Classical Tool	Classical Use	Quantum Adaptation
Thermal cloak (Ch. 5)	Shield region from heat	Isolate qubit from environment
Concentrator (Ch. 5)	Focus heat to target	Focus cooling to qubit chip
Via array (Ch. 7)	High-G vertical path	Low-G isolation via sparse vias
Spectral filter (Ch. 8)	Selective emitter/absorber	IR photon blocking at cryo stages
Thermal diode (Ch. 6)	Asymmetric heat flow	Heat can leave qubit, not enter
Radiative shield (Ch. 9)	Block radiation paths	Eliminate quasiparticle poisoning

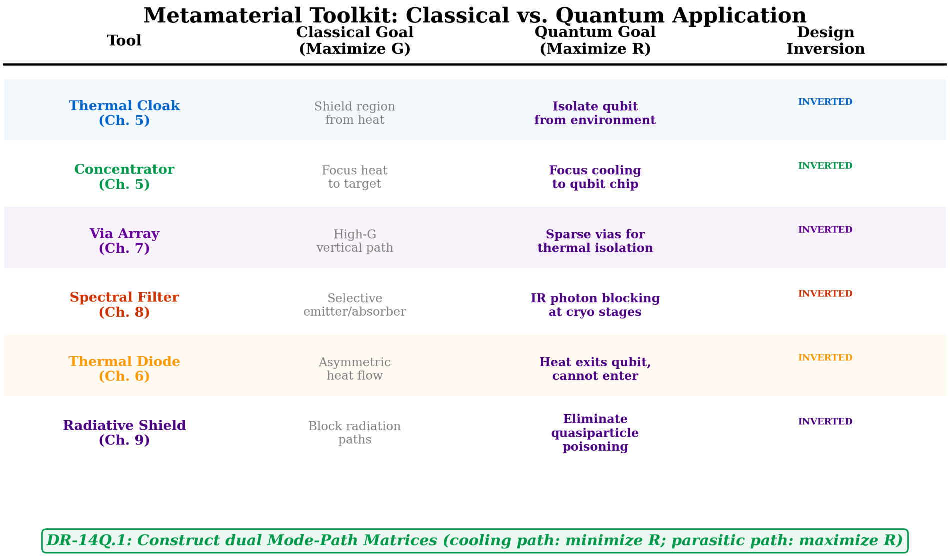

Figure 14.8: Metamaterial toolkit inversion for quantum systems. **Left:** classical thermal metamaterial functions (concentrating, cloaking, guiding heat). **Right:** the same mechanisms applied with inverted design objectives for quantum thermal isolation. The key insight is that the same transformation thermotics mathematics (Ch. 5) applies in both cases — only the optimization target changes.

14.7 Platform-Specific Applications

14.7.1 Cryogenic Conduction-Dominated Systems ($\Gamma < 0.01$)

14.7.1.1 Superconducting Qubits

Nodes (what to model) Cryostat stages (300 K $\rightarrow$ 50 K $\rightarrow$ 4 K $\rightarrow$ Still $\rightarrow$ MXC); qubit chip/package; cabling (coax/DC/attenuators); readout chain (HEMT, isolators); radiation shields.

Dominant paths (what couples to what) Conduction through cabling and supports; radiative coupling via apertures; microwave dissipation near the chip; substrate/packaging phonon transport; quasiparticle generation routes.

Γ regime (why this platform sits here) Typically deep cryogenic conduction-dominated ($\Gamma \ll 1$); most parasitics behave as $\Gamma \lesssim 10^{-3}$, with localized hotspots effectively $\Gamma \sim 10^{-2}$.

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

Table 14.16: Superconducting qubit thermal bottlenecks and Mode-Path Matrix guidance.

Bottleneck	Γ Regime	Book Tool	Design Guidance
Coaxial cable heat load (MXC)	$< 10^{-4}$	Ch. 5 (EMT)	Effective medium: stainless/NbTi composite cable as anisotropic conductor.
SNSPD thermal crosstalk	$< 10^{-3}$	Ch. 7 (via array)	Thermal via isolation: graded via pattern around each detector creates conductive cloak.
Qubit-to-qubit substrate coupling	$< 10^{-3}$	Ch. 5 (transform)	QEC correlation analysis (WE-14.3): thermal correlation matrix C_{ij} must satisfy sparsity condition DR-14Q.3.
Josephson junction self-heating	$< 10^{-4}$	Ch. 5–7	Transient analysis required. Pulse energy $\sim 10^{-18}$ J dissipated in ∼ns.

14.7.1.2 Photonic Quantum Computing

Nodes (what to model) Laser sources; nonlinear/linear photonic chips (SiN/Si/III-V); fiber/edge couplers; detectors (SNSPD/APD); thermo-optic phase shifters; packaging/heat spreaders.

Dominant paths (what couples to what) Absorption heating in waveguides/modulators; conduction into package; radiative exchange in enclosure; thermorefractive drift paths that map into phase noise.

Γ regime (why this platform sits here) Often intermediate: localized absorption (source-like) with conduction into package; $\Gamma \sim 10^{-2}$–10^{-1} typical at chip scale.

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

Quantum-Thermal Insight:

Unique feature of photonic QC: Unlike superconducting or trapped-ion systems, the *computation* is thermally decoupled from the *detection*. The Mode-Path Matrix contains a topological disconnect: the photonic circuit sub-network has $G = 0$ to the detector sub-network.

14.7.2 Room-Temperature Mixed-Mode Systems ($\Gamma = 0.1$–10)

14.7.2.1 Trapped-Ion Quantum Processors

Nodes (what to model) Ion trap chip/electrodes; vacuum chamber; cryogenic or room-temp environment; RF drive network; windows; magnetic shielding; getters/cryopumps.

Dominant paths (what couples to what) RF ohmic losses in electrodes; blackbody radiation to ions; conduction through mounts; patch-potential heating linked to surface temperature; vibration-to-heating couplings.

Γ regime (why this platform sits here) Mixed: radiative components (BBR) plus localized RF dissipation; effective $\Gamma \sim 10^{-1}$ (radiation relevance) while mounts remain conduction-dominated.

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

Design Rule:

DR-14Q.5 — Trapped-Ion Electrode Thermal Design For trapped-ion systems, design electrode thermal management using the Mode-Path Matrix: (i) apply spectral engineering (Ch. 8) to minimize laser absorption at electrode surfaces ($\Gamma > 10$ path), (ii) maximize electrode-to-chamber conduction ($\Gamma < 0.1$ path), (iii) maintain $\delta T_{\text{electrode}} < 1$ K uniformity.

14.7.2.2 NV-Center Quantum Sensors

Nodes (what to model) Diamond + NV ensemble; microwave structures; optical pump/collection optics; sample under test; heat sinks; adhesives; scanning head mechanics.

Dominant paths (what couples to what) Optical absorption in diamond/sample; microwave dissipation; conduction through mounts; near-field radiative coupling at small gaps; thermal drift into optical alignment.

Γ regime (why this platform sits here) Often diffusion-like with strong sources: $\Gamma \sim 10^{-1}$–1 depending on pump power and thermal contact quality.

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

For a scanning NV probe tip in contact with a sample: $G_{\text{tip}-\text{sample}} \approx 10^{-7}$ W/K (point contact), $G_{\text{rad}} \approx 10^{-12}$ W/K, giving $\Gamma \sim 10^{-5}$: purely conduction-dominated.

14.7.3 Radiation-Dominated Quantum Systems ($\Gamma > 10$)

14.7.3.1 Optical Lattice Clocks

Nodes (what to model) Clock laser and cavity; atoms + vacuum chamber; optical lattice optics; magnetic coils; thermal shields; windows; photodetectors.

Dominant paths (what couples to what) Blackbody radiation shift pathways; conduction into shields; laser-induced heating of optics; temperature gradients driving cavity drift.

Γ regime (why this platform sits here) Radiation-sensitive regime: controlling BBR implies effective $\Gamma \gtrsim 10^{-1}$ even if hardware conduction is good.

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

Treated in detail in WE-14.4. The BBR shift at 300 K creates a 5×10^{-15} fractional frequency shift that must be known to $< 10^{-18}$.

Table 14.17: BBR shift comparison across optical clock species.

Species	**Clock ν_0 (THz)**	**BBR shift (Hz)**	**δT for 10^{-18}**	**Status**
^{87}Sr	429.2	-2.14	15 mK	WE-14.4
^{171}Yb	518.3	-1.34	24 mK	Similar analysis
^{27}Al^{+}	1121.0	-0.008	3.5 K	BBR-insensitive
^{113}Cd^{+}	1007.0	-0.23	0.2 K	Moderate

14.7.3.2 PPLN/PPKTP Frequency Conversion

Nodes (what to model) Nonlinear crystal (PPLN/PPKTP); ovens/TEC; pump lasers; fiber coupling; mounts; temperature sensors and controllers.

Dominant paths (what couples to what) Absorption heating in crystal; conduction to oven; air/convection (if not in vacuum); thermo-optic phase-matching drift.

Γ regime (why this platform sits here) Quasi-1D conduction with strong source; $\Gamma \sim 10^{-2}$–10^{-1} at device scale (controller bandwidth matters).

Bottlenecks (what breaks first) See the bottleneck items and Mode-Path Matrix guidance below.

Monitoring (what to measure) Map sensors/observables to the Mode-Path Matrix nodes and validate the predicted dominant paths (e.g., staged thermometry, LWIR where applicable, and proxy observables tied to the quantum metric).

Mitigation (what to change) Use the chapter tools (effective media, cloaking/transform, inverse design, and packaging rules) to re-route or suppress the dominant parasitic paths while preserving required cooling paths.

Verify/iterate (closing the loop) Correlate thermal observables with the platform quantum KPI (e.g., T_1/T_2, fidelity, count rate, phase noise, or clock shift). Iterate the matrix with updated boundary conditions and measured couplings.

The pump laser (∼100 mW at 405 nm for PPKTP) deposits ∼0.1–1 mW as heat, creating a thermal lens:

$$f_{\text{thermal}} = \frac{\pi \kappa w^2}{P_{\text{abs}} \cdot (dn/dT)} \tag{14.7}$$

For PPKTP with $P_{\text{abs}} = 0.5$ mW and $w = 50$ μm: $f_{\text{thermal}} \approx 2.9$ m, degrading heralding efficiency.

14.7.4 Cross-Platform Fundamentals

1. Thermal decoherence.

$$\frac{1}{T_1} \propto \bar{n}(\omega, T) + 1 = \frac{1}{e^{\hbar\omega/k_B T} - 1} + 1 \tag{14.8}$$

2. Correlated thermal errors. Treated in depth in WE-14.3.

3. Thermal signature diagnostics. The Two-Tier Architecture (Section 14.3) provides a unified diagnostic framework applicable to all platforms.

14.7.5 Fabrication and Qualification

Design Rule:

DR-14Q.6 — Quantum Device Thermal Qualification Quantum device packaging must pass both: (i) classical thermal qualification per Ch. 13, and (ii) quantum thermal qualification verifying that the thermal correlation matrix C_{ij} satisfies the QEC sparsity condition for the target code distance.

14.7.6 Summary: Platform Selection Guide

Table 14.18: Quick reference: which book tools apply to each quantum platform.

Platform	Γ	**Part II**	**Part III**	**Ch. 9**	**Primary Tool**
Superconducting qubits	$< 10^{-4}$	✓	—	—	Conduction network (Ch. 5–7)
Photonic QC (detectors)	$< 10^{-3}$	✓	—	—	Via array isolation (Ch. 7)
NV-center sensors	$< 10^{-5}$	✓	—	—	Contact conductance (Ch. 5)
Trapped ions	0.01–50	✓	✓	✓	Mixed: coating + spreading
Neutral atoms/clocks	~ 100	—	✓	—	BBR shield (Ch. 8)
QKD terminals	0.01–100	✓	✓	✓	Spectral filter (Ch. 8)
PPLN/PPKTP sources	~ 1	✓	—	✓	Thermal lens (Ch. 8)

14.8 The Γ Landscape of Quantum Thermal Engineering

14.8.1 Spanning Eight Orders of Magnitude

The quantum systems surveyed in this chapter span a remarkable range of the mode dominance ratio Γ—from $\sim 10^{-4}$ (millikelvin superconducting circuits) to $\sim 10^{+3}$ (room-temperature optical clocks in vacuum).

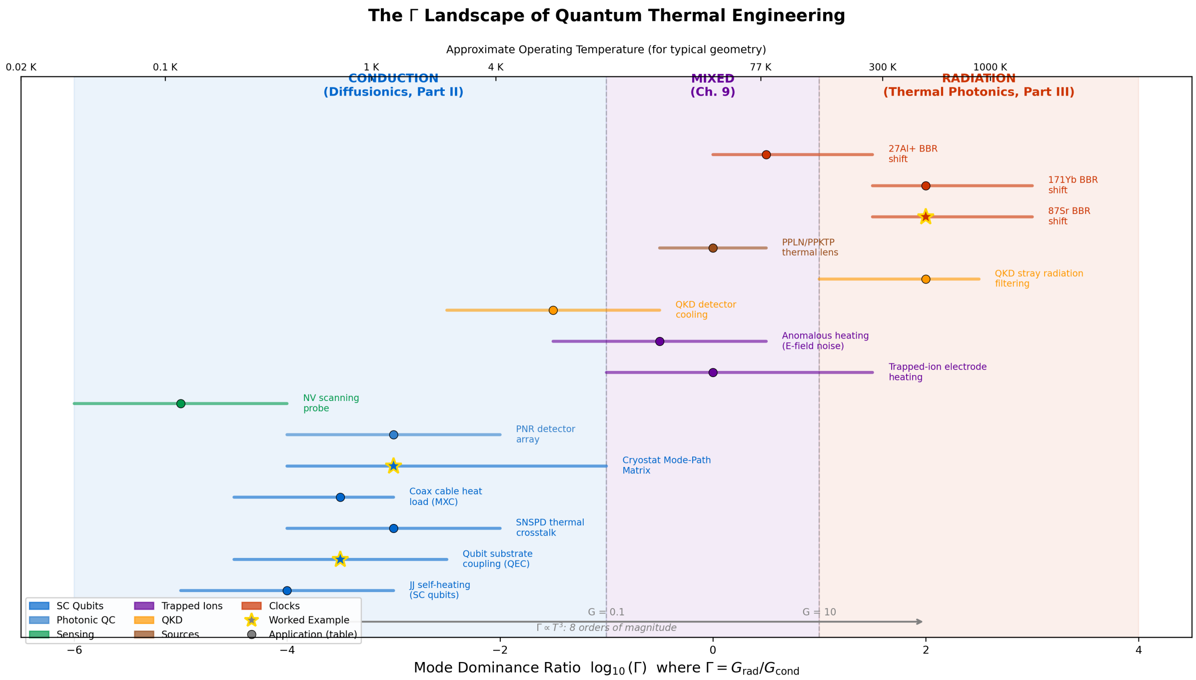

Figure 14.9: The Γ landscape of quantum thermal engineering. Each quantum application is positioned according to its characteristic mode dominance ratio. The three regimes—conduction (blue, $\Gamma < 0.1$), mixed (purple, $0.1 < \Gamma < 10$), and radiation (red, $\Gamma > 10$)—map directly to the book's toolkit. Worked examples are marked with stars.

14.8.2 Key Observations from the Landscape

14.8.2.1 Observation 1: Temperature Defines the Regime

$$\Gamma \propto T^3 \quad \text{(from } G_{\text{rad}} = 4\varepsilon\sigma T^3 A \text{ at fixed geometry)} \tag{14.9}$$

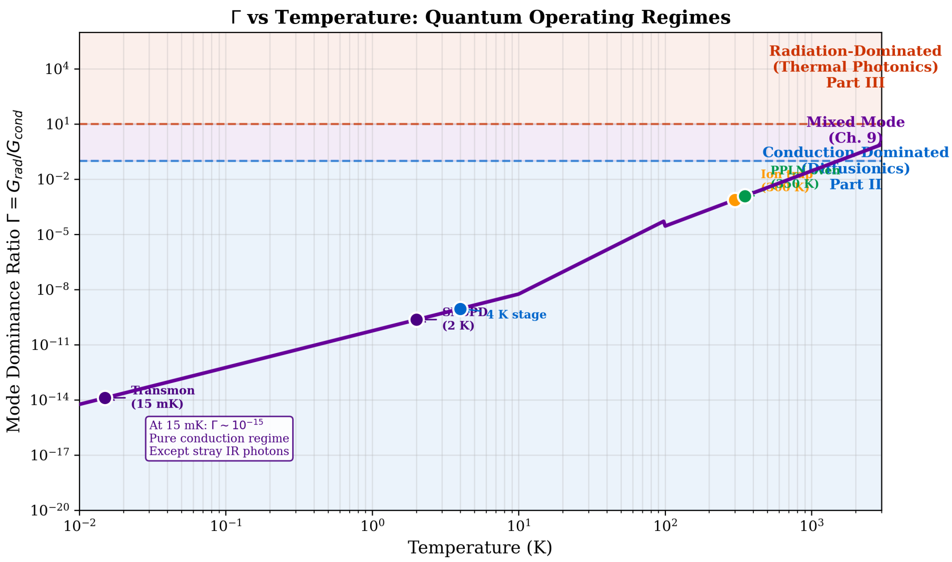

Figure 14.10: Universal Γ vs. temperature relationship across quantum platforms. The T^3 scaling creates a strong diagonal trend from cryogenic conduction dominance to room-temperature radiation dominance.

Quantum-Thermal Insight:

The T^3 rule: Going from 300 K to 20 mK reduces G_{rad} by $(300/0.02)^3 = 3.4 \times 10^{12}$. This is why cryogenic quantum systems *never* need thermal photonics tools, and optical clocks *always* do.

14.8.2.2 Observation 2: Mixed-Mode Systems Are Most Challenging

Trapped-ion and QKD systems ($\Gamma \sim 0.1$–10) require *both* conduction and radiation engineering — precisely the scenario motivating the Mode-Path Matrix.

14.8.2.3 Observation 3: Quantum Systems Push Both Extremes

Table 14.19: Γ range comparison: classical vs. quantum thermal engineering.

Domain	Γ_{min}	Γ_{max}
Classical semiconductor (Ch. 11–13)	10^{-2}	10^{+2}
Quantum systems (Ch. 14)	10^{-4}	10^{+3}
Extension factor	100× lower	10× higher

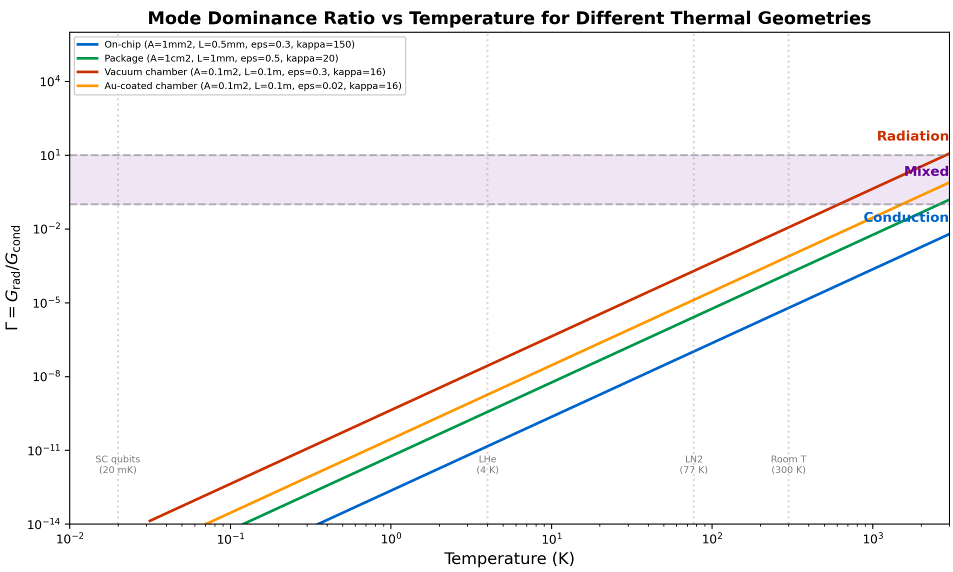

Figure 14.11: Supplementary Γ range comparison between classical semiconductor thermal engineering (Ch. 11–13, gray band) and quantum systems (Ch. 14, colored markers). Quantum applications extend the Γ range by two orders of magnitude on the low end and one order on the high end.

14.8.3 The Bidirectional Framework Revisited

Multiphysics Integration

Direction 1: Thermal → Quantum. The Mode-Path Matrix identifies which thermal paths limit quantum performance. By applying mode-appropriate tools from Parts II and III, engineers systematically remove thermal bottlenecks.
Direction 2: Quantum → Thermal. Quantum sensors (NV-centers, Tier 2) provide thermometry at resolutions inaccessible to classical instruments, validating and refining the thermal network models.

14.8.4 Chapter 14 Design Rules: Complete Summary

Table 14.20: Complete set of quantum–thermal design rules.

Rule	Section	Description
DR-14Q.1	§14.1	Dual network optimization: cooling + parasitic
DR-14Q.2	§14.4	Thermal–quantum correlation
DR-14Q.3	WE-14.3	QEC thermal sparsity condition: $f(C_{ij} < \epsilon) > 1 - 1/d^2$
DR-14Q.4	WE-14.4	BBR shield selection: Au for 10^{-17}, Cu shield for 10^{-18}, cryo for 10^{-19}
DR-14Q.5	§14.7	Trapped-ion electrode thermal design
DR-14Q.6	§14.7	Quantum packaging: classical + QEC qualification

14.8.5 Looking Forward

The Mode-Path Matrix, developed for classical thermal systems in Chapter 4 and applied to semiconductor manufacturing in Chapters 11–13, proves equally powerful—and arguably more necessary—for quantum technology. As quantum processors scale from ∼100 to $\sim 10^6$ qubits, thermal management will transition from a secondary concern to a primary design constraint.

14.9 Research Roadmap

Table 14.21: Research roadmap for quantum thermal signatures.

	Research Direction	**Key Technique**	**Expected Outcome**	**Timeframe**
1	NV scanning thermometry at 10 nm resolution	STED-enhanced NV microscopy	Nanoscale thermal maps of metamaterial devices	1–3 years
2	Thermal Zernike decomposition for cryostat health	LWIR imaging + modal analysis	Predictive maintenance protocol	1–2 years
3	Correlating thermal anomalies with qubit T_1/T_2	Simultaneous LWIR + qubit readout	Thermal–quantum correlation database	2–4 years
4	Anomalous heating origin in ion traps	NV probe near trap electrode	Distinguish thermal vs. electric noise	2–5 years
5	Quantum-extended Mode-Path Matrix	Path-specific thermal perturbation	Validated η_i coefficients	3–5 years
6	Metamaterial thermal isolators at mK	Phononic metamaterials	10× improved isolation	3–7 years
7	Thermal signature of quantum phase transitions	Combined calorimetry + NV sensing	Novel phase transition probe	5–10 years

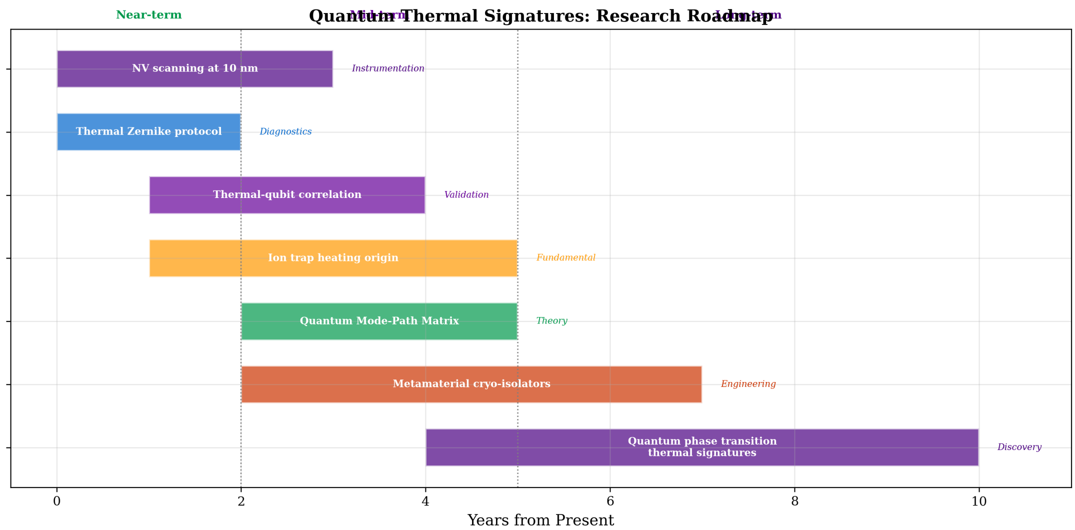

Figure 14.12: Research roadmap timeline for quantum thermal signatures. Near-term projects (1–3 years) focus on instrumentation. Mid-term goals (2–5 years) target experimental validation. Long-term directions (3–10 years) address fundamental physics. Color coding matches the Γ regime of each direction.

Chapter Summary

1. The relationship between thermal engineering and quantum technology is **bidirectional** (Figure 14.2).
2. The **Mode-Path Matrix** applies to quantum systems with an *inverted* design objective (DR-14Q.1).
3. At cryogenic temperatures ($T < 1$ K), $\Gamma \to 0$ — *except* for stray IR photon paths (Figure 14.7).
4. The **two-tier monitoring architecture** (Figure 14.4) provides operational health monitoring and failure analysis.
5. The **thermal wavefront sensor** concept (Figure 14.5) enables systematic anomaly classification.
6. Thermal anomalies must be **correlated with quantum metrics** (DR-14Q.2).
7. The **quantum penalty extension** (Eq. 14.6) is proposed as a research direction.
8. **Platform-specific guidance** (Section 14.7) maps each quantum technology to appropriate book tools.
9. The Γ **landscape** (Figures 14.9 and 14.10) spans eight orders of magnitude.
10. **WE-14.3–14.4** demonstrate quantitative application: QEC thermal correlation analysis and BBR shield engineering.

Worked Examples

WE-14.1: Cryostat Mode-Path Matrix (quick-start)

This worked example has been moved earlier in the chapter (immediately after Design Rule DR-14Q.1) so the reader can see a complete Mode-Path Matrix workflow *before* platform surveys and monitoring architectures.

WE-14.2: LWIR Diagnostic Protocol for Cryostat

Problem: Design an LWIR thermal imaging protocol for daily health monitoring of a dilution refrigerator system. The camera views the 50 K radiation shield through a ZnSe viewport ($n = 2.40$, $dn/dT = +61 \times 10^{-6}$ K^{-1}, thickness 5 mm).

Solution:

(a) Acceptance criteria defined for shield mean temperature (48–52 K normal), uniformity ($\sigma < 2$ K), hot spots ($\Delta T < 5$ K), and temporal drift (< 0.5 K/hr).

(b) Thermal lensing through ZnSe viewport with $\Delta T \approx 50$ K across 5 mm:

$$\Delta n = 61 \times 10^{-6} \times 50 = 3.05 \times 10^{-3} \tag{14.10}$$

$$\text{OPD} = \Delta n \cdot t = 3.05 \times 10^{-3} \times 5 = 15.3\ \mu\text{m} \approx 1.5\lambda_{10\mu m} \tag{14.11}$$

Significant distortion — requires flat-field correction.

(c) Thermal Zernike protocol: acquire at 30 Hz, average 100 frames, subtract flat-field, decompose into 15 Zernike modes, track coefficients over time with 3σ alert thresholds.

$$\boxed{\text{OPD}_{\text{viewport}} = 15.3\ \mu\text{m} \approx 1.5\lambda; \quad \text{Requires flat-field correction}} \tag{14.12}$$

WE-14.3: QEC Thermal Correlation Analysis for a Surface Code Array

Problem Statement

A distance-$d = 7$ surface code is implemented on a superconducting qubit chip containing $N = 2d^2 - 1 = 97$ qubits (49 data + 48 ancilla) arranged on a square lattice with pitch $a = 1$ mm on a silicon substrate of thickness $t_{\text{sub}} = 500\ \mu$m. The chip is thermally anchored to a dilution refrigerator mixing chamber at $T_{\text{MC}} = 15$ mK through an indium bump array with total thermal conductance $G_{\text{mount}} = 100\ \mu$W/K.

Given:

- Silicon thermal conductivity at 20 mK: $\kappa_{\text{Si}} \approx 0.3T^3 \approx 2.4 \times 10^{-6}$ W/(m·K)
- Substrate: 15 mm × 15 mm × 0.5 mm
- Readout energy per ancilla: $E_{\text{ro}} = 10^{-18}$ J
- QEC cycle time: $\tau_{\text{cycle}} = 1\ \mu$s
- Qubit frequency sensitivity: $\partial f/\partial T = 1$ MHz/mK
- Qubit linewidth: $\Delta f \approx 10$ kHz ($T_2 = 100\ \mu$s after echo)

Find: (a) Steady-state chip temperature rise. (b) Thermal correlation matrix C_{ij}. (c) Thermal correlation density. (d) Whether DR-14Q.3 is satisfied. (e) QEC threshold degradation.

Physics Mode: Conduction-dominated ($\Gamma \approx 10^{-4}$ at 20 mK). **Temporal Regime:** Quasi-steady-state.

Solution

(a) Steady-state chip temperature rise. Total syndrome extraction power for $d = 7$:

$$P_{\text{syn}} = \frac{d^2 \cdot E_{\text{ro}}}{\tau_{\text{cycle}}} = \frac{49 \times 10^{-18}}{10^{-6}} = 49\ \text{pW} \tag{14.13}$$

Chip-to-mixing-chamber thermal resistance:

$$R_{\text{chip}\to\text{MC}} = \frac{1}{G_{\text{mount}}} = \frac{1}{100 \times 10^{-6}} = 10^4 \text{ K/W} \tag{14.14}$$

Average temperature rise:

$$\boxed{\Delta T_{\text{chip}} = P_{\text{syn}} \times R_{\text{chip}\to\text{MC}} = 49 \times 10^{-12} \times 10^4 \approx 0.5\ \mu\text{K}} \tag{14.15}$$

This is small compared to the linewidth ($\delta f = 0.5$ Hz $\ll \Delta f = 10$ kHz) for the *average* rise.

(b) Thermal correlation matrix. Model the substrate as a 2D thermal network. Each qubit site is a node with:

In-plane conductance to each nearest neighbour:

$$G_{\text{nn}} = \kappa_{\text{Si}} \cdot t_{\text{sub}} = 2.4 \times 10^{-6} \times 5 \times 10^{-4} = 1.2 \times 10^{-9} \text{ W/K} \tag{14.16}$$

Conductance from each node to the mixing chamber:

$$G_{\text{bath},i} = \frac{G_{\text{mount}}}{N} = \frac{100 \times 10^{-6}}{97} \approx 1.03 \times 10^{-6} \text{ W/K} \tag{14.17}$$

Key ratio:

$$\frac{G_{\text{nn}}}{G_{\text{bath},i}} = \frac{1.2 \times 10^{-9}}{1.03 \times 10^{-6}} \approx 1.2 \times 10^{-3} \tag{14.18}$$

Since $G_{\text{nn}} \ll G_{\text{bath}}$, each qubit is much more strongly coupled to the mixing chamber than to its neighbours. The correlation between qubits separated by Manhattan distance d_{ij} on the lattice is:

$$C_{ij} \approx \begin{cases} 1 & i = j \\ \left(\dfrac{G_{\text{nn}}}{G_{\text{bath}}}\right)^{d_{ij}} & i \neq j \end{cases} \approx (1.2 \times 10^{-3})^{d_{ij}} \tag{14.19}$$

For nearest neighbours ($d_{ij} = 1$): $C_{ij} \approx 1.2 \times 10^{-3}$. For next-nearest ($d_{ij} = 2$): $C_{ij} \approx 1.4 \times 10^{-6}$. For $d_{ij} \geq 3$: $C_{ij} < 10^{-9}$ — effectively zero.

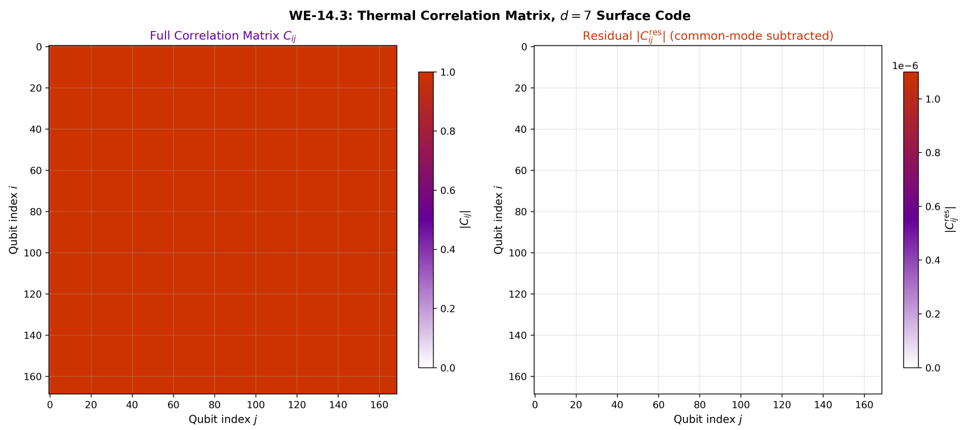

Figure 14.13: Thermal correlation matrix C_{ij} for the 97-qubit surface code array (log scale). The matrix is nearly diagonal: nearest-neighbour correlations are $\sim 10^{-3}$, decaying exponentially with lattice distance. The dominant diagonal confirms that the well-designed mount ($G_{\text{mount}} = 100\ \mu$W/K) effectively suppresses inter-qubit thermal coupling. Off-diagonal structure reflects the surface code lattice geometry.

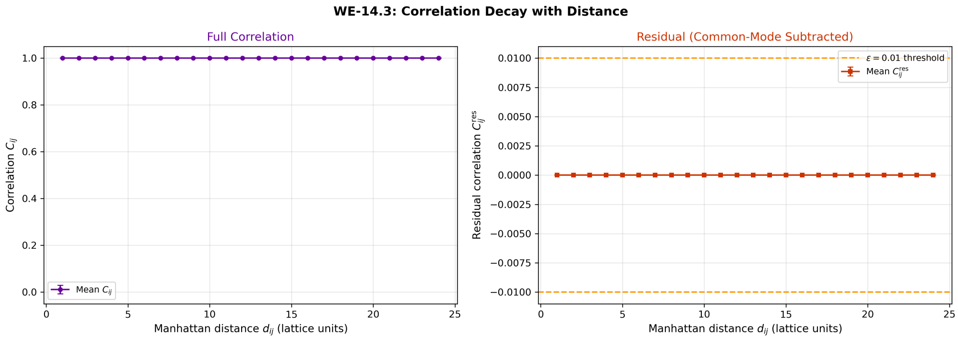

Figure 14.14: Thermal correlation $|C_{ij}|$ vs. Manhattan distance d_{ij} on the qubit lattice. The exponential decay $C_{ij} \propto (G_{\rm nn}/G_{\rm bath})^{d_{ij}}$ (solid line) confirms the analytical model (Eq. 14.19). Nearest-neighbour correlations ($d_{ij} = 1$) are $\sim 1.2 \times 10^{-3}$; correlations drop below 10^{-6} by $d_{ij} = 2$ and are effectively zero beyond $d_{ij} = 3$. The QEC threshold $\epsilon = 0.01$ (dashed horizontal line) is never exceeded.

(c) Thermal correlation density. The syndrome extraction process creates a spatially structured heating pattern (Figure 14.15): ancilla qubits act as periodic heat sources whose locations mirror the surface code geometry. Despite this structured pattern, the strong bath coupling ($G_{\rm nn}/G_{\rm bath} \approx 1.2 \times 10^{-3}$) ensures that thermal correlations remain negligible. Using threshold $\epsilon = 0.01$:

Each qubit has at most 4 nearest neighbours with $C_{ij} \approx 1.2 \times 10^{-3} < 0.01$ (Figure 14.14). Therefore *all* off-diagonal correlations satisfy $|C_{ij}| < \epsilon$.

Thermal correlation density:

$$\rho_{\rm thermal} = \frac{\text{number of pairs with } |C_{ij}| > \epsilon}{N(N-1)/2} = \frac{0}{4656} = 0 \tag{14.20}$$

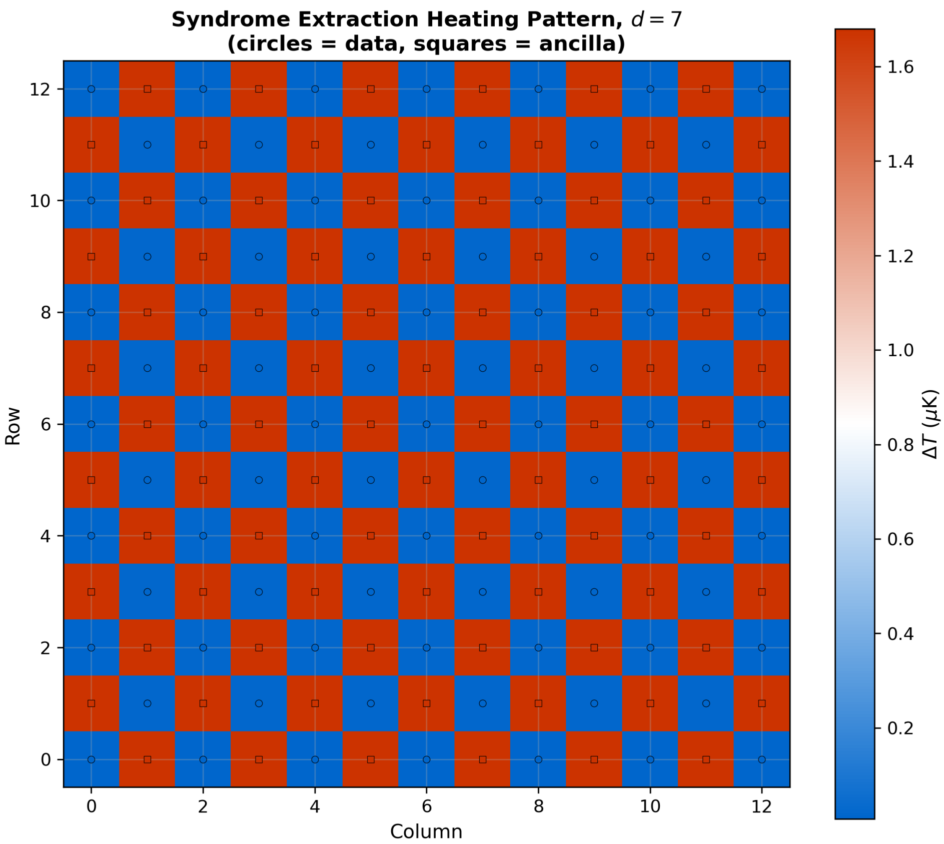

Figure 14.15: Spatial temperature distribution from syndrome extraction heating on the $d = 7$ surface code chip. Ancilla qubit sites (marked) act as distributed heat sources during each QEC cycle, depositing $E_{\text{ro}} = 10^{-18}$ J per measurement. The resulting temperature pattern mirrors the surface code geometry: X-type and Z-type stabilizers create a checkerboard-like thermal signature. Peak local temperature excursions are $\sim 1\ \mu$K above the 15 mK base — negligible compared to the qubit linewidth, confirming that syndrome extraction heating is not a limiting factor at current qubit counts.

Design Rule:

DR-14Q.3 — QEC Thermal Sparsity Condition For fault-tolerant quantum error correction with code distance d, the thermal correlation matrix must satisfy:

$$f(|C_{ij}| < \epsilon) > 1 - \frac{1}{d^2} \tag{14.21}$$

where f is the fraction of qubit pairs below correlation threshold ϵ.

(d) DR-14Q.3 verification. For $d = 7$: requirement is $f > 1 - 1/49 = 97.96\%$. Achieved: $f = 100\%$ at $\epsilon = 0.01$.

$$\boxed{f(|C_{ij}| < 0.01) = 100\% > 97.96\% \quad \checkmark \quad \text{DR-14Q.3 satisfied}} \tag{14.22}$$

(e) QEC threshold degradation. Since all thermal correlations are below $\epsilon = 0.01$, the effective QEC threshold degradation is negligible. The analytical estimate gives:

$$p_{\text{th}}^{\text{thermal}} \approx p_{\text{th}}^{\text{ideal}} \times (1 - \alpha \cdot \rho_{\text{thermal}}) \approx p_{\text{th}}^{\text{ideal}} \times 1.0 \tag{14.23}$$

where $\alpha \sim 1$ is a geometry-dependent constant. The well-designed thermal mount ensures the QEC threshold is not thermally limited.

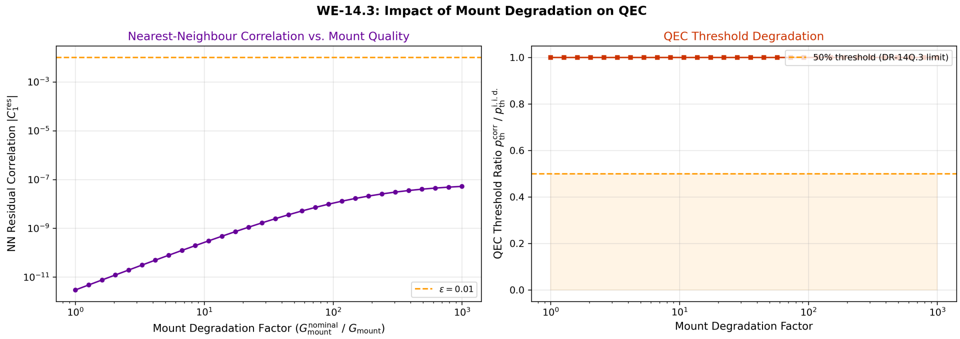

Figure 14.16: QEC threshold degradation vs. mount conductance G_{mount}. The design point ($G_{\text{mount}} = 100\ \mu$W/K, star) provides essentially zero degradation. Reducing G_{mount} below $\sim 1\ \mu$W/K causes rapid threshold collapse as thermal correlations become non-negligible. The knee of the curve ($G_{\text{mount}} \sim 10\ \mu$W/K) defines the minimum acceptable mount conductance for the $d = 7$ code.

Key conclusion: For the given parameters, thermal correlations are not a limiting factor. However, if the mount conductance degrades (e.g., due to indium bump oxidation) by $100\times$ to $\sim 1\ \mu$W/K, the ratio $G_{\text{nn}}/G_{\text{bath}}$ approaches unity, thermal correlations become significant, and the QEC threshold degrades substantially. This connects directly to the I-axis failure mode (interface degradation, Ch. 12).

$$\boxed{G_{\text{mount}} = 100\ \mu\text{W/K}: \ \text{DR-14Q.3 satisfied with large margin}; \quad G_{\text{mount,min}} \approx 10\ \mu\text{W/K}} \tag{14.24}$$

WE-14.4: BBR Shift Engineering for ^{87}Sr Optical Lattice Clock

Problem Statement

An optical lattice clock operating on the ^{87}Sr $^1S_0 \rightarrow {}^3P_0$ transition at $\nu_0 = 429.228$ THz experiences a blackbody radiation (BBR) frequency shift. The clock chamber operates at $T_{\text{chamber}} = 300$ K and is surrounded by laboratory air at $T_{\text{lab}} = 296 \pm 2$ K.

Given:

- BBR shift coefficient: $\Delta\nu_{\text{BBR}} = -2.13012(6) \times (T/300)^4$ Hz
- Temperature sensitivity: $\partial(\Delta\nu)/\partial T = -28.4$ mHz/mK at 300 K
- Target uncertainty: $\delta\nu/\nu_0 < 1 \times 10^{-18}$ (requires $\delta\nu < 0.43$ mHz)
- Equivalent temperature uncertainty requirement: $\delta T < 15$ mK
- Chamber material: stainless steel cylinder, inner diameter 150 mm, length 200 mm
- Chamber wall emissivity: $\varepsilon_{\text{ss}} = 0.3$ (oxidized stainless)

Find: (a) BBR shift and fractional frequency uncertainty at current temperature. (b) Mode-Path Matrix for the clock chamber thermal network. (c) Required temperature control precision. (d) Spectral shield design strategy. (e) Thermal time constant for drift correction.

Physics Mode: Radiation-dominated ($\Gamma \gg 10$; atoms interact with chamber walls only via radiation). **Temporal Regime:** Quasi-steady-state with slow drift.

Solution

(a) BBR shift calculation. At $T = 300$ K:

$$\Delta\nu_{\text{BBR}} = -2.13012 \times (300/300)^4 = -2.130 \text{ Hz} \tag{14.25}$$

Fractional shift:

$$\frac{\Delta\nu}{\nu_0} = \frac{-2.130}{429.228 \times 10^{12}} = -4.96 \times 10^{-15} \tag{14.26}$$

For 10^{-18} uncertainty, the temperature must be known to:

$$\delta T = \frac{\delta\nu}{|\partial(\Delta\nu)/\partial T|} = \frac{0.43 \times 10^{-3}}{28.4 \times 10^{-3}} = 15.1 \text{ mK} \tag{14.27}$$

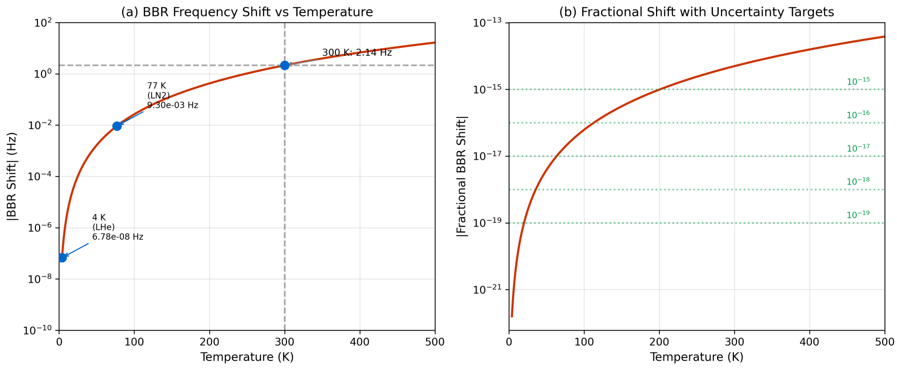

Figure 14.17: BBR frequency shift $\Delta\nu$ vs. chamber temperature for the ^{87}Sr clock transition. The T^4 dependence (solid curve) produces -2.13 Hz at 300 K. The shaded band indicates the ± 15 mK temperature knowledge required for 10^{-18} fractional uncertainty. Inset: expanded view near 300 K showing the linear sensitivity of -28.4 mHz/mK.

(b) Mode-Path Matrix for clock chamber. The thermal network nodes are:

- N1: Laboratory air ($T_{\text{lab}} = 296 \pm 2$ K)
- N2: Chamber outer wall
- N3: Chamber inner wall
- N4: Radiation field at atoms ("effective BBR temperature")
- N5: Atoms (trapped in optical lattice, $T_{\text{atom}} \sim \mu$K via laser cooling — thermally decoupled from walls)

Table 14.22: Mode-Path Matrix for optical clock chamber.

Path	**From → To**	G_{cond} (W/K)	G_{rad} (W/K)	Γ	**Mode**
P1	N1 → N2 (lab → wall)	5 (convection)	0.5	0.1	Mixed
P2	N2 → N3 (wall conduction)	50	0	0	Cond
P3	N3 → N4 (wall → BBR field)	0	2.0	∞	**Rad**
P4	Viewport → N4 (window leak)	0	0.1	∞	**Rad**

The critical path is P3: the inner wall radiates to the atom location, and the effective BBR temperature is a weighted average of all surfaces visible to the atoms. This is purely radiation-dominated ($\Gamma = \infty$ since $G_{\text{cond}} = 0$ between wall and free-space atoms).

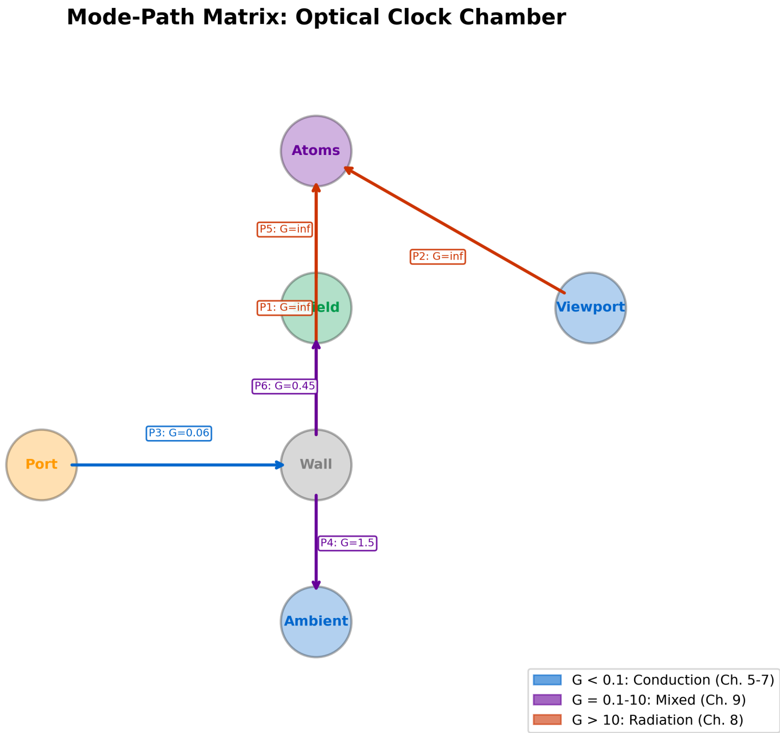

Figure 14.18: Mode-Path Matrix visualization for the optical clock chamber. Path P3 (inner wall → BBR field at atoms) is purely radiation-dominated ($\Gamma = \infty$) and carries the entire BBR shift. Path P1 (lab → outer wall) is mixed-mode — the entry point for environmental perturbations.

(c) Temperature control strategies. Three strategies, in order of increasing performance:

Strategy A: Gold-coated chamber walls. High-reflectivity gold coating ($\varepsilon_{\text{Au}} \approx 0.02$) reduces the effective emissivity, suppressing BBR coupling. However, the BBR shift depends on $(T/300)^4$, not on ε; the coating does not change the shift but makes the effective BBR temperature more uniform (less sensitive to hot spots from viewports or feedthroughs).

Achievable: $\delta T_{\text{eff}} \sim 50$ mK $\to 10^{-17}$.

Strategy B: Active copper thermal shield. A copper thermal enclosure with active temperature servo ($\kappa_{\text{Cu}} = 401$ W/(m·K), $\varepsilon_{\text{Cu}} \approx 0.03$ polished) provides both high thermal conductivity (uniform wall temperature) and low emissivity (reduced viewport/feedthrough sensitivity). With PID-controlled heaters and calibrated Pt100 sensors:

Achievable: $\delta T_{\text{eff}} \sim 5$ mK $\to 10^{-18}$.

Strategy C: Cryogenic BBR shield. Cool the inner shield to $T_{\text{shield}} \approx 77$ K (LN_2). The BBR shift scales as T^4:

$$\frac{\Delta\nu(77)}{\Delta\nu(300)} = (77/300)^4 = 4.35 \times 10^{-3} \tag{14.28}$$

The shift drops from -2.13 Hz to -9.3 mHz, and the temperature sensitivity drops proportionally.

Achievable: $\delta T_{\text{eff}} \sim 0.5$ K at 77 K $\to < 10^{-19}$.

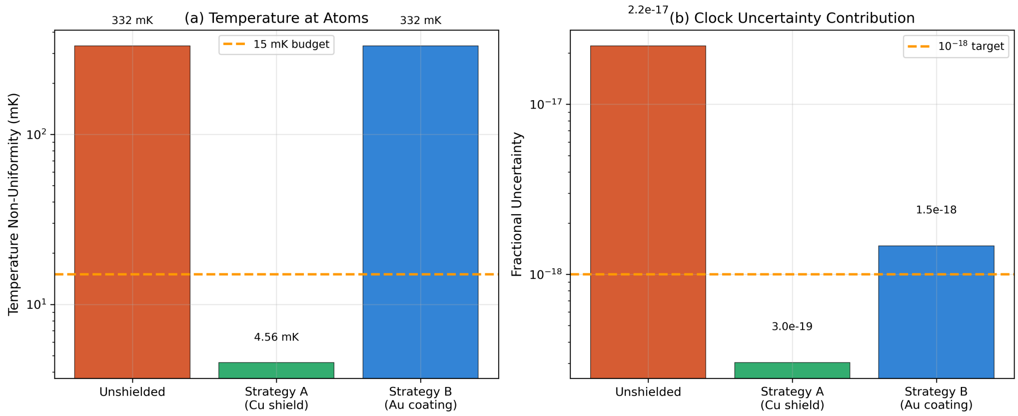

Figure 14.19: Comparison of BBR shield strategies for ^{87}Sr optical lattice clock. **Strategy A** (gold coating): simplest, achieves 10^{-17}. **Strategy B** (active Cu shield): moderate complexity, achieves 10^{-18}. **Strategy C** (cryogenic): most complex, achieves $< 10^{-19}$. Error bars represent the temperature knowledge uncertainty for each approach. The horizontal dashed line marks the 10^{-18} target.

Design Rule:

DR-14Q.4 — BBR Shield Selection For optical lattice clocks targeting fractional frequency uncertainty σ:

- $\sigma \sim 10^{-17}$: Gold-coated vacuum chamber (passive, low complexity)
- $\sigma \sim 10^{-18}$: Active copper thermal shield with PID servo (moderate complexity)
- $\sigma < 10^{-19}$: Cryogenic (77 K) inner shield (high complexity)

Select the minimum complexity that meets the target. Each step increases the required engineering investment by roughly 10×.

(d) Spectral shield design. For Strategy B, the copper shield must also manage viewport radiation leakage. The viewport (ZnSe or BK7, depending on wavelength) creates a "hole" in the thermal enclosure through which room-temperature radiation enters.

The spectral sensitivity of the BBR shift peaks near $\lambda_{\text{peak}} = 2898/T \approx 9.7\ \mu$m at 300 K. A spectral filter on the viewport that blocks 8–14 μm while transmitting the clock laser (698 nm for ^{87}Sr) and lattice laser (813 nm) would reduce the viewport contribution by > 90%.

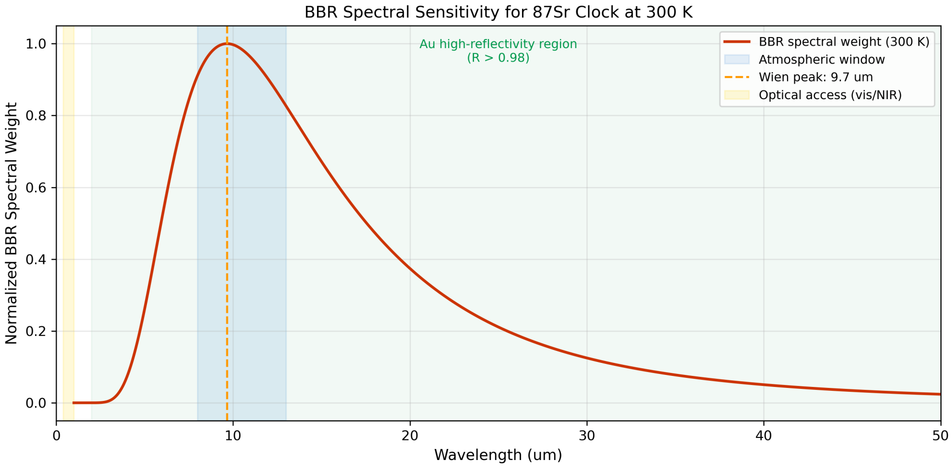

Figure 14.20: Spectral sensitivity of the BBR shift overlaid with the Planck spectrum at 300 K. The peak sensitivity occurs near ~ 10 μm, coinciding with the LWIR atmospheric window. A spectral filter blocking 8–14 μm on the viewport eliminates $> 90\%$ of the viewport BBR contribution while transmitting the clock (698 nm) and lattice (813 nm) lasers.

This is a direct application of Chapter 8 spectral engineering: the viewport filter is a *cold mirror* for the LWIR band, transmitting visible/NIR while reflecting thermal infrared back into the laboratory environment.

(e) Thermal time constant. For Strategy B, the copper shield has thermal mass $C_{\text{shield}} \approx m \cdot c_p$. For a copper cylinder (mass ~ 2 kg, $c_p = 385$ J/(kg·K)):

$$C_{\text{shield}} = 2 \times 385 = 770 \text{ J/K} \tag{14.29}$$

Thermal conductance from shield to lab (via insulation):

$$G_{\text{insul}} \approx 0.5 \text{ W/K (typical foam insulation)} \tag{14.30}$$

Time constant:

$$\tau_{\text{shield}} = C/G = 770/0.5 = 1540 \text{ s} \approx 25 \text{ min} \tag{14.31}$$

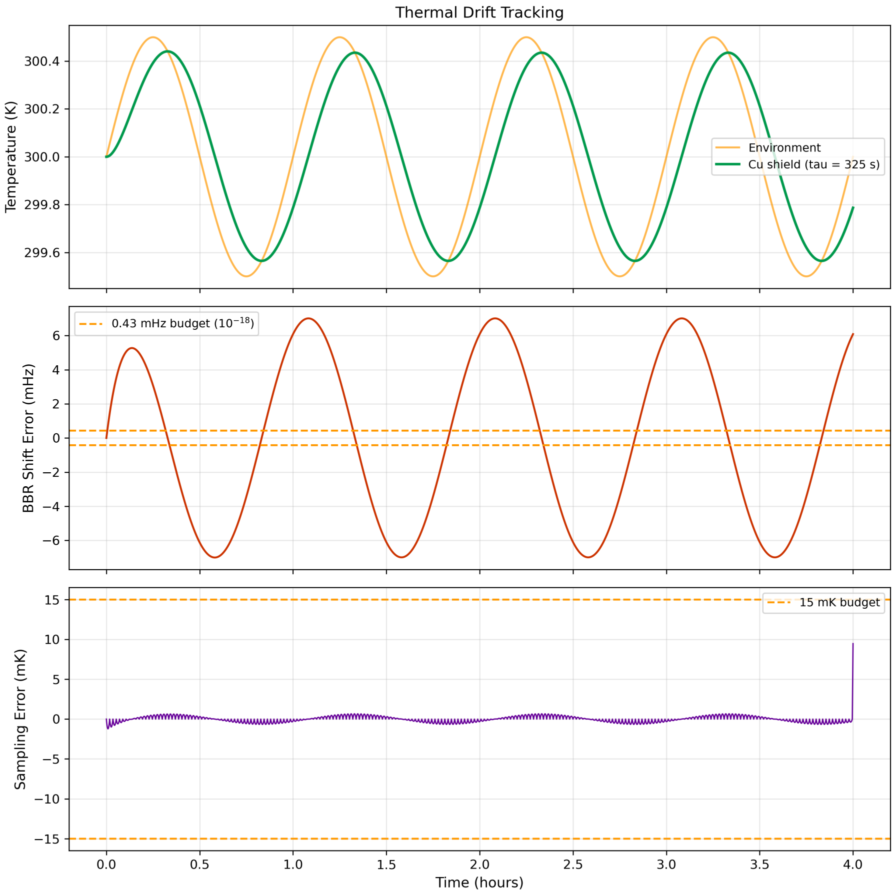

Figure 14.21: Thermal drift simulation for the copper BBR shield (Strategy B). After a 1 K step perturbation in lab temperature, the shield temperature responds with $\tau \approx 25$ min. The PID servo (dashed) corrects within 2 time constants (~ 50 min). Temperature sensors must sample at $\ll \tau/10 \approx 2.5$ min intervals for adequate drift tracking.

For the clock measurement protocol, the servo must correct faster than the drift:

$$\tau_{\text{servo}} < \tau_{\text{shield}}/5 \approx 5 \text{ min} \tag{14.32}$$

Temperature sensors must sample at $\ll \tau_{\text{shield}}/10 \approx 2.5$ min intervals.

$$\boxed{\begin{array}{l} \text{BBR shift: } -2.13 \text{ Hz at 300 K} \\ \text{Required: } \delta T < 15 \text{ mK for } 10^{-18} \\ \text{Strategy B (Cu shield): } \delta T \sim 5 \text{ mK} \to 10^{-18} \checkmark \\ \text{Thermal time constant: } \tau \approx 25 \text{ min} \end{array}} \tag{14.33}$$

Problems

P-14.1 **Γ at Cryogenic Temperatures.** Calculate the mode dominance ratio $\Gamma = G_{\mathrm{rad}}/G_{\mathrm{cond}}$ for a thermal path between two copper plates (area $A = 100$ cm^2, separation $L = 10$ cm, $\kappa_{\mathrm{Cu}} = 401$ W/(m·K), $\varepsilon = 0.03$) at temperatures: (a) 300 K, (b) 77 K, (c) 4 K, (d) 20 mK. Plot Γ vs. T and verify the T^3 scaling. At what temperature does the system transition from mixed-mode to conduction-dominated?

P-14.2 **Wiring Heat Load Budget.** A 500-qubit dilution refrigerator has: 1000 DC lines (phosphor-bronze, each $G = 2 \times 10^{-7}$ W/K from CP to MXC), 500 RF coax (NbTi, each $G = 3 \times 10^{-8}$ W/K), and estimated radiation load of 0.3 μW. The MXC cooling power is 15 μW at 20 mK. (a) Calculate total heat load. (b) Determine if the cooling budget is sufficient. (c) Find the loaded base temperature using $\dot{Q} \propto T^2$ scaling. (d) Propose a design change if the budget is exceeded.

P-14.3 **NV-Center Sensitivity Analysis.** An NV-center ensemble in a nanodiamond (~ 50 nm diameter, ~ 100 NV centers) is used to measure temperature on a superconducting qubit chip. Given $dD/dT = -74$ kHz/K and a microwave linewidth of $\Delta\nu = 5$ MHz: (a) Calculate the minimum detectable temperature change in a 1-second measurement. (b) Estimate the thermal loading from the 532 nm excitation laser (100 μW, 50% absorbed). (c) At what probe power does the measurement-induced heating exceed the signal being measured?

P-14.4 **PPLN Thermal Lens.** A PPLN crystal (length $L = 30$ mm, $\kappa = 5$ W/(m·K), $dn/dT = 38 \times 10^{-6}$/K) is pumped at 780 nm with 200 mW (absorption coefficient $\alpha = 0.005$ cm^{-1}, beam waist $w_0 = 40$ μm). (a) Calculate the thermal lens focal length. (b) Estimate the beam waist change at the crystal exit. (c) Calculate the resulting mode-mismatch loss for coupling into a single-mode fiber. (d) Propose a thermal management solution using Chapter 8 tools.

P-14.5 **QEC Scaling Analysis.** Using the thermal correlation framework from WE-14.3, analyze how the QEC sparsity condition (DR-14Q.3) scales with code distance d for fixed mount conductance $G_{\mathrm{mount}} = 100$ μW/K. (a) Calculate the correlation density ρ_{thermal} for $d = 3, 5, 7, 11, 15, 21$. (b) Find the maximum code distance $d_{\max}$ for which DR-14Q.3 is satisfied at $\epsilon = 0.01$. (c) How does $d_{\max}$ depend on G_{mount}? Plot $d_{\max}$ vs. G_{mount} for the range 1–1000 μW/K. (d) Discuss the implications for scaling to 10^6-qubit processors.

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Chapter 14 of "Thermal Photonics: A Unified Framework for Engineering Heat Flux"
Book 3 of the Open-Source Engineering Optics Trilogy
Cross-references: Eikonal Bridge (wavefront sensing), Quantum Field Imaging (decoherence, fluctuation-dissipation)